



IDMP TEST SERIES - 2022

High Yielding PRELIMS TEST SERIES

Test – 19

Environment & Ecology

Topics covered:

- Ecology, Functions Of An Ecosystem
- Terrestrial Ecosystem Aquatic Ecosystem
- Environmental Pollution
- Renewable Energy
- Environmental Issues
- Current Affairs January to June 2021



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DEDICATED PRELIM PROGRAM(DPP)

UPSC-CSE PRELIMINARY TEST SERIES – 2021/2022

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TEST BOOKLET SERIES

Test Code:

DPP

TEST BOOKLET

GENERAL STUDIES

TEST -19

E19

Time Allowed: Two Hours

Maximum Marks : 200

INSTRUCTIONS

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC.

2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE A, B, C OR D AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.

3. You have to enter your Roll Number on the Test Booklet in the provided alongside this line. DO NOT write anything else on the Test Booklet.

4. This Test Booklet contains 100 questions. Each question is printed in English. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose ONLY ONE response for each item.

5. All items carry equal marks.

6. Penalty for wrong answers:

THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.

- I. There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, One-Third of the marks assigned to that question will be deducted as penalty.
- II. If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question.
- III. If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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TOPICS COVERED: Ecology, Functions Of An Ecosystem, Terrestrial Ecosystem Aquatic Ecosystem, Environmental Pollution, Renewable Energy, Environmental Issues, Current Affairs January to June 2021



Q.1) With reference to the Ecological Pyramid, consider the following statements :

1. The Base of the Ecological Pyramid is occupied by the food producers while Tip is occupied by the carnivores.
2. The number, biomass and energy of organisms gradually increase with each step/trophic level of the pyramid

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.2) With reference to Ecological Pyramids, Which of the following pairs is Incorrect ?

- a) Forest Ecosystem - An Upright Pyramid of Number
- b) Grassland Ecosystem - An Inverted Pyramid of Number
- c) Aquatic Ecosystem - An Inverted Pyramid of Biomass
- d) Grassland Ecosystem - An Upright Pyramid of Biomass

Q.3) With reference to Biofertilizers, consider the following statements :

1. Biofertilizer is a substance which contains living microorganisms.
2. Biofertilizers add nutrients through the natural processes of fixing atmospheric nitrogen by action of bacteria.
3. Biofertilizers decrease phosphorus uptake by solubilizing and releasing unavailable phosphorus.

Which of the statements given above is/are Correct ?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.4) With reference to Estuaries, consider the following Statements :

1. An estuary is a coastal body of water where flow of freshwater from rivers mixes with saltwater of the ocean.
2. Estuaries are homes to plants and animals which have low salt tolerance.
3. Estuaries create a natural barrier that absorbs the energy of the waves and prevents them from flooding cities.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.5) With reference to Ramsar Convention, consider the following statements :

1. It is an intergovernmental treaty that provides the framework for the conservation and wise use of wetlands and their resources
2. The Administrative services to the convention are provided by Bird life International.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 & 2
- d) Neither 1 nor 2

Q.6) With reference to Ecological Adaptation, consider the following statements :

1. Presence of higher amount of Red Blood Cells in Tribes living at high altitude in Himalayas is an example of Physiological Adaptation.
2. The acquired effects of adaptation pass on to the next generation.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.7) With reference to Ecological Niche, consider the following statement :

1. Ecological niche is the functional role and position of species in its ecosystem.
2. Ecosystem stability has an inverse relation with Niche Diversity.
3. The ecological niche of dominant species is broader and more extensive compared to that of less dominant species.

Which of the above statements is/are correct ?

- a) 1 and 2 only
- b) 1 and 3 only
- c) 3 only
- d) 1, 2 and 3

Q.8) Which of the following are the adaptations found in mangroves?

1. Viviparity mode of reproduction
2. Stilt roots
3. Pneumatophores



- 4. Thin leaves
- 5. Lenticellated bark

Select the correct answer using the code given below:

- a) 1, 2, 3 and 5 only
- b) 1, 2, 4 and 5 only
- c) 2, 3 and 4 only
- d) 1, 2, 3, 4 and 5

Q.9) Which of the following is not the criteria for identification of wetlands of national importance under the Ramsar Convention?

- a) If a site regularly supports 20,000 or more water birds.
- b) If a site regularly supports 10% of the individuals in a population of one species or subspecies of waterbirds.
- c) If a site supports a significant proportion of indigenous fish subspecies, species or families
- d) If a site supports vulnerable, endangered, or critically endangered species; or threatened ecological communities.

Q.10) Which of the following statements with respect to India State of Forest Report, 2019 is correct?

- a) All States in the Northeast show decrease in forest cover.
- b) The mangrove cover in the country has decreased by 540 sq. km as compared to the previous assessment.
- c) The total carbon stock in the country's forests has increased as compared to the last assessment of 2017.
- d) Area wise Chhattisgarh has the largest forest cover in the country.

Q.11) Consider the following statements with reference to Wetlands (Conservation and Management) Rules, 2017:

- 1. The rules are based on decentralization in management of wetlands.
- 2. They provide for setting up the State Wetlands Authority and National Wetlands Committee under the chairmanship of State Environment Minister and Union Environment Minister respectively.
- 3. The rules are applicable to wetlands categorized as 'wetlands of national importance' under the Ramsar convention only.

Which of the above statements is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3 only

Q.12) Which of the following adaptations can be found in the desert plants?

- 1. Reduced size or absence of leaves
- 2. Presence of chlorophyll in the stems of the plants
- 3. Narrower root system
- 4. Reduction in photosynthesis during dry period

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 2, 3 and 4 only
- c) 1, 2 and 4 only
- d) 1, 3 and 4 only

Q.13) Which of the following statements with reference to cold water corals is correct?

- a) They do not need sunlight for their survival.
- b) Because of cold conditions they are only small in size.
- c) Cold water corals grow much faster than the warm water corals.
- d) None of the above.

Q.14) Consider the following statements with reference to the Grassland ecosystem in India:

- 1. These grasslands are found where rainfall is 25-75cm per year.
- 2. In western parts of Rajasthan, mainly savanna grassland ecosystems are found.
- 3. 'Vidi' is a form of grassland ecosystems of Gujarat.

Which of the above statements is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) 2 and 3 only

Q.15) Consider the following statements with reference to Kelp forests:

- 1. Kelp forests are large brown algae found in coastal areas.
- 2. They require cold, nutrient-rich and relatively clear water.
- 3. They show a very high rate of growth, as high as half a meter a day.

Which of the above statements are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.16) Consider the following statements with reference to eutrophication:



1. It is mainly caused by the addition of nitrates and phosphates into the aquatic ecosystem.
2. It promotes anaerobic conditions in the aquatic ecosystem.
3. The growth of green algae in the lake surface layer is physical identification of eutrophication.

Which of the above statements are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.17) These types of forests are found along the Western Ghats, the Nicobar and Andaman Islands and all along the northeastern region. These are characterized by tall, straight trees. The more common trees in these forests are the jackfruit, betel nut palm, jamun, and mango. The trees in these forests form a tier pattern: shrubs cover the layer closer to the ground, followed by the short-structured trees and then the tall variety.

Which of the following forest types is described in the above passage?

- a) Tropical dry evergreen forests
- b) Tropical dry deciduous forests
- c) Tropical wet evergreen forests
- d) Tropical moist deciduous forests

Q.18) Consider the following statements with reference to swamp ecosystems:

1. These are wetlands marked by the complete absence of trees.
2. They are characterized by mineral soils.
3. They exist in areas with poor drainage and sufficient water supply.

Which of the above statements is/are incorrect?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1 and 3 only

Q.19) Consider the following statements with reference to the estuary ecosystem:

1. In estuaries, water level and salinity rise and fall with the tides and seasons.
2. Flow of salt water into the estuaries is called estuarine current, which is warmer than anti-estuarine currents.

Which of the above statements is/are correct?

- a) 1 only

- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.20) Consider the following pairs of type of forest and region in which they are primarily found in India:

Type of Forests	Region
1. Littoral and Swamp forests	— Andaman and Nicobar
2. Tropical thorn forests	— Gujarat
3. Sub alpine forests	— Arunachal Pradesh
4. Montane wet temperate forests	— Andhra Pradesh

Which of the above pairs are correctly matched?

- a) 1, 2 and 3 only
- b) 1 and 4 only
- c) 2 and 3 only
- d) 3 and 4 only

Q.21) Consider the following pairs of Ramsar sites in India and their location:

Ramsar Site	Location
1. Kolleru lake	— Andhra Pradesh
2. Ashtamudi	— Tamil Nadu
3. Lonar lake	— Maharashtra
4. Kabartal	— Bihar

Which of the above pairs are correctly matched?

- a) 1, 2 and 3 only
- b) 2, 3 and 4 only
- c) 1, 3 and 4 only
- d) 1, 2 and 4 only

Q.22) Consider the following ecosystems:

1. Temperate forests
2. Savanna
3. Tundra
4. Estuaries

Which of the following is the correct sequence in decreasing order of mean net primary productivity of the given ecosystems?

- a) 1-4-2-3
- b) 2-4-1-3
- c) 4-1-2-3
- d) 3-2-1-4

Q.23) Consider the following statements with reference to Temperate desert in India:



1. These are found in Ladakh, Spiti valley of Himachal Pradesh and Sikkim.
2. They are arid because of the rain-shadow effect of the Himalayas.
3. Soil in these deserts has low water retention capacity.

Which of the above statements is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3 only

Q.24) Consider the following statements with reference to desertification:

1. In India, desertification is confined to Rajasthan, Gujarat, Punjab and Haryana due to the presence of Aravalis.
2. The United Nations Convention to Combat Desertification is the sole legally binding international agreement linking environment and development to sustainable land management.
3. India has pledged to restore 26 million hectares of degraded land by 2025.

Which of the above statements is/are correct?

- a) 1 and 2 only
- b) 2 only
- c) 2 and 3 only
- d) 1 and 3 only

Q.25) Consider the following pollutants:

1. Phenol
2. Silica
3. Glycerol
4. Carbohydrates
5. Mercury

Which of the above pollutants are inorganic in nature?

- a) 1, 3, and 5 only
- b) 2 and 5 only
- c) 2, 3, and 5 only
- d) 1, 2, 3, 4, and 5 only

Q.26) Consider the following statements:

1. Decomposition of organic matter increases the Dissolved Oxygen (DO) of the water.
2. Chemical Oxygen Demand (COD) lowers the concentration of DO in water.
3. The concentration of DO is varied by respiration and photosynthesis only.

Which of the statements given above is/are correct?

- a) 1 and 3 only

- b) 1 and 2 only
- c) 2 only
- d) 1 only

Q.27) Consider the following factors:

1. Use of combine harvesters
2. Paddy transplantation
3. Intensification of crops
4. Short duration rice cultivation
5. Crop diversification

Which of the factors mentioned above are responsible for the burning of stubble by the farmers?

- a) 1, 2, 3, and 5 only
- b) 1 and 3 only
- c) 1, 3, 4, and 5 only
- d) 2, 3, and 5 only

Q.28) Consider the following statements:

1. Bioaugmentation is an ex-situ bioremediation technique that enhances the degradation process.
2. Bioremediation works efficiently on all kinds of degradable and non-degradable wastes.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.29) In the context of the Environmental Pollution (Prevention and Control) Authority, consider the following statements:

1. It is authorized to take suo moto cognizance of cases relating to environmental pollution.
2. It deals with the implementation of the Graded Response Action Plan and the National Clean Air Programme.
3. SAFAL, SWAS, STAR are its initiatives to minimize the air pollution around the NCR region.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 1 only
- c) 2 only
- d) 2 and 3 only

Q.30) Consider the following statements:

1. Instances of acid rain are not found in the mountainous regions with thin soils.
2. To qualify for acid rain, the pH value of the rainwater must be less than 2.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only



- c) Both 1 and 2
d) Neither 1 nor 2

Q.31) In the context of natural and anthropogenically acidified lakes, consider the following statements:

1. The concentration of dissolved carbon is high in the former while low in the latter.
2. The transparency of water is lower in case of naturally acidified lakes while high in case of the latter.
3. Both are very well buffered and are abundant with different species and rich aquatic life.

Which of the statements given above is/are correct?

- a) 1 and 2 only
b) 2 and 3 only
c) 1 and 3 only
d) 1, 2, and 3

Q.32) Consider the following statements:

1. Parthenocarpic fruits are produced without the formation of seeds.
2. Both alcoholic and non-alcoholic beverages compulsorily contain at least one hydroxyl group.

Which of the statements given above is/are correct?

- a) 1 only
b) 2 only
c) Both 1 and 2
d) Neither 1 nor 2

Q.33) Consider the following pollutants:

1. Sulphur dioxide
2. Lead
3. Aluminum
4. Phosphoric acid
5. Methyl Mercury
6. Cadmium

Which of the pollutants given above are triggered with the acid rain and are not directly associated with it?

- a) 1, 2, 4, and 5 only
b) 2, 3, 5, and 6 only
c) 1, 3, 4, and 6 only
d) 2, 4, 5, and 6 only

Q.34) Which of the following are the likely consequences for the aquatic organisms if the increase in the temperature of the water body?

1. Increased enzyme activity.
2. Increased photosynthetic activity.
3. Increased metabolic rate of organisms.
4. Increase in the population of Blue green algae.

5. Increased migration of organisms.

Select the correct answer using the codes given below:

- a) 1, 3, and 5 only
b) 1, 2, 3, 4, and 5
c) 3, 4, and 5 only
d) 1, 2, and 4 only

Q.35) Consider the following statements regarding plastic pollution:

1. The Ultraviolet light induced degradation of plastics is more pronounced on land than in marine environments.
2. Plastics are traditionally considered to be bio-inert with no toxic effect on aquatic life directly.

Which of the statements given above is/are correct?

- a) 1 only
b) 2 only
c) Both 1 and 2
d) Neither 1 nor 2

Q.36) Consider the following pairs:

Waste treatment process

1. Sanitary landfills
2. Vermiculture bacteria
3. Pyrolysis without oxygen
4. Incineration under controlled oxygen

Feature

Leaching
Fungi and

Burning

Burning

Which of the pairs given above is/are incorrectly matched?

- a) 1, 2, and 3 only
b) 1, 2, and 4 only
c) 1, 3, and 4 only
d) 2, 3, and 4 only

Q.37) Which of the statements given below is least appropriate in relation to composting?

- a) Composting is considered good for the growth of plants.
- b) Composting is the degradation of organic wastes carried out by microorganisms.
- c) Composting is done either in the absence of oxygen or under controlled levels of oxygen.
- d) The moisture retaining capability of the soil is enhanced post the composting process.



Q.38) Consider the following statements:

1. Gamma radiation is regarded as the most severe to human health.
2. Ultraviolet radiations are an example of short-wave nonionizing radiations.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.39) Consider the following statements:

1. The Central Ground Water Authority and the Central Water Commission are created under the purview of the Environment (Protection) Act, 1986.
2. India is a signatory to the International Convention for the Prevention of Pollution from ships.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.40) In relation to E-Waste (Management) Rules, 2016, which of the statement(s) given below is least appropriate?

1. The rules do not cover the electrical components like Mercury based CFLs.
2. It envisages State wise authorization of the Extended Producer Responsibility.
3. The rules are applicable to both the producers as well as the consumers of the product.

Select the correct answer using the codes given below:

- a) 1 and 2 only
- b) 1 only
- c) 2 and 3 only
- d) 1, 2, and 3

Q.41) Which of the following are an example of the Waste to Wealth systems?

1. Mixing of ash residue in Plasma Arc gasification with cement.
2. Pyrolysis oil obtained from the heating of the polymers.
3. Production of syngas in the biogas digester.

Select the correct answer using the codes given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.42) Consider the following statements:

1. Organochlorines are an example of Persistent Organic Pollutants (POPs).
2. The regulation of POPs in India is done under the Environment (Protection) Act, 1986.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.43) Consider the following regarding Hydrogen Fuel cells:

1. A constant source for charging is required.
2. Cost efficient due to lower cost of the catalysts.
3. Generates electricity by way of a chemical reaction.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.44) Consider the following statements:

1. Setting up of Compressed Biogas Plants is an important part of SATAT.
2. One sun, one World, One Grid focused on European and American states in the first phase.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.45) Consider the following:

1. Longer charging time.
2. Reduced noise and cheaper maintenance.
3. Higher cost of the vehicles.
4. Short distance travels.

Which of the above are hindrances in the adoption of Electric Vehicles in India?

- a) 1, 3, and 4 only
- b) 1 and 3 only
- c) 2, 3, and 4 only
- d) 2 and 4 only



Q.46) Which of the following plant mechanisms is influenced by light ?

1. Chlorophyll production.
2. Vegetative development.
3. Effect on transpiration rate.
4. Stomatal movements.
5. Distribution of plants.

Select the correct answer from the codes given below

- a) 1, 2, 3 and 4 only.
- b) 1, 3 and 5 only
- c) 2, 3, 4 and 5 only
- d) 1, 2, 3, 4 and 5.

Q.46) With respect to the pyramid of energy, consider the following statements.

1. It is also known as the trophic pyramid and is always upright.
2. Concept of ecological efficiency can be explained with its help.
3. Shorter the food chain more will be food energy available.

Which of the given statements above is/are correct ?

- a) 1 and 2 only
- b) 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.47) Match the following lists regarding various ecological rules.

List 1 (Rules)

List 2 (Related)

A- Bergmann's Rule : Body size with respect to warm/cold weather condition

B- Allen's Rule : Body surface area to volume ratio.

C- Rensch's Rule : Sexual size dimorphism.

D- Jordan's Rule : Temperature effect on morphology.

E- Gloger's Rule : Pigmentation changes with respect to environments.

Select the correct answer from the codes given below:

- a) A-1, B-2, C-3, D-4, E-5
- b) A-2, B-1, C-3, D-5, E-4
- c) A-2, B-1, C-5, D-4, E-3
- d) A-1, B-3, C-2, D-4, E-5

Q.48) Arrange the following level of organization of an ecosystem in ascending order starting from protoplasm.

1. Organisms
2. Tissues
3. Population
4. Ecosystem
5. Cells
6. Communities
7. Biomes

Select the correct answer from the codes given below:

- a) $5 < 2 < 1 < 3 < 6 < 4 < 7$
- b) $5 < 2 < 1 < 6 < 3 < 4 < 7$
- c) $2 < 5 < 1 < 6 < 3 < 7 < 4$
- d) $2 < 5 < 1 < 3 < 6 < 7 < 4$

Q.49) With respect to the ecological niche of r-strategists, consider the following statements.

1. They have large demands owing to their need for specific conditions.
2. Reproduce rapidly under favorable conditions.

Which of the above given statements is/are correct ?

- 1 only
- 2 only
- Both 1 and 2
- Neither 1 nor 2

Q.50) Principle of reciprocal exclusion suggests:

- a) Two species that possess the same ecological niche cannot live together in the same ecosystem.
- b) Each species having its own vital preference can develop very well within its adaptive potentialities, taking into accounts competitions.
- c) Species do not express its entire genetic potential. It develops only as part of its potential.
- d) Both b and c

Q.51) Which of the following statements regarding circadian rhythm is/are correct?

1. It is a response mechanism due to various environmental factors .
2. Observed only on birds and mammals and is a vital part of their survival .

Select the correct answer from the codes given below :

- a) only
- b) only
- c) Both 1 and 2
- d) Neither 1 nor 2



Q.52) Which of the following is/are the structural factors of the ecosystem?

1. Species composition
2. Stratification
3. Trophic Structure
4. Standing state
5. Standing crop

Select the correct answer from the codes given below :

- a) 1, 2 and 4 only
- b) 3 only
- c) 3, 4 and 5 only
- d) 1, 2, 3, 4 and 5

Q.53) Diapause can be defined as,

- a) Period of an organism spending a specific time of the year in a dormant stage.
- b) A period of suspended development during unfavorable conditions.
- c) Prolonged dormancy period of an organism during a hot or dry period.
- d) A period of non activity during cell cycle and cell division of an organism.

Q.54) With respect to ecology, the term resource partitioning means,

- a) Division of limited resources by species to help avoid competition.
- b) Immigration in maintaining the persistence of the local population that would otherwise go extinct.
- c) When both species superficially resemble each other.
- d) Decline in production under conditions of low population density.

Q.55) Which of the following statements is incorrect regarding Biomass:

- a) Pyrolysis, gasification are the processes that convert Biomass into usable forms of energy.
- b) Biomass is a renewable resource that does not add any extra carbon to the atmospheric balance sheet.
- c) India's renewable energy target includes 10 GW of renewable capacity from Bio-power.
- d) Burning of Biomass does not release any carbon dioxide into the atmosphere unlike fossil fuels.

Q.56) Consider the following statements:

- 1) The National Clean Air Programme (NCAP) aims to achieve the 20% to 30% reduction in Particulate Matter concentration by 2034.

- 2) It is prepared by the Ministry of Housing and Urban affairs to reduce PM concentration in the selected smart cities.

Select the correct answer from the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.57) Consider the following statements:

- 1) Biological Oxygen Demand (BOD) is a better indicator to measure the concentration of required oxygen as compared to Chemical Oxygen Demand (COD).
- 2) COD measures the concentration of oxygen required for the oxidation of non-biodegradable matters while BOD measures the same for the degradable matter.
- 3) The dissolved oxygen is inversely proportional to the dissolved organic wastes in the water body.

Select the correct answer from the codes given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1, 2 and 3
- d) None of the above

Q.58) Which of the following statements are incorrect regarding the Solid Wastes Management Rules, 2016?

- a) Waste generators are to segregate wastes into three categories: biodegradable, non-biodegradable, and domestic hazardous wastes.
- b) Local authorities and village panchayats are to make suitable waste management plans in accordance with the state government policy.
- c) All manufacturers of disposable products are to provide financial assistance to the local authorities for the setting up of waste disposal systems.
- d) These rules are only applicable to the plains and exempt the hilly and the north eastern states.

Q.59) Consider the following statements regarding Bioremediation:

- 1) It uses microorganisms or their enzymes to break down pollutants from the environment.
- 2) Marine Microbes Wheat Bran is a hydrocarbon degrading bacteria with the capacity to clear oil spills.
- 3) These microbes do not have a metabolism of their own and rely on the hydrocarbon waste for energy.



Select the correct answer from the codes given below:

- a) 1 and 2 only
- b) 1 only
- c) 1 and 3 only
- d) 2 and 3 only

Q.60) With reference to Terrestrial Ecosystems, consider the following statements-

- 1. Temperature and Precipitation are the two limiting factors of the Terrestrial Ecosystems.
- 2. Plants are at the base of the food web in the terrestrial ecosystems.
- 3. Alterations and changes in land use is one of the major concerns of the terrestrial ecosystem.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above.

Q.61) In the context of Terrestrial and Aquatic Ecosystems, consider the following statements-

- 1. Grazing food webs are common to both Terrestrial and Aquatic Ecosystems.
- 2. Both terrestrial and aquatic ecosystems are affected by temperature, salinity, and sunlight.
- 3. Terrestrial ecosystems can support more trophic levels than the aquatic ecosystems.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 2 only
- d) 2 only

Q.62) With reference to Aquatic Ecosystems, consider the following statements-

- 1. Changes in precipitation alters the hydrologic characteristics of aquatic systems.
- 2. The level of the Dissolved oxygen in aquatic ecosystems is a function of salinity.
- 3. Aphotic Zone is where the environmental factors are conducive for photosynthesis.

Which of the statements given above is/are incorrect?

- a) 1 and 3 only
- b) 2 only
- c) 3 only
- d) 2 and 3 only

Q.63) Consider the following statements-

- 1. Estuaries are essential components of the Terrestrial Ecosystem.
- 2. Monocropping is a threat to the Grassland Ecosystems.
- 3. Fungi acts as decomposers in aquatic environments.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 2 only

Q.64) Consider the following pairs-

Terms	Some associated Characteristics
1. Eutrophication	The nutrient-enrichment of the lakes
2. Limnetic Zone	Sunlight does not penetrate here.
3. Trophic cascade	The addition or removal of top predators in an ecosystem.

Select the correct answer using the code below-

- a) 2 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1 only

Q.65) With reference to Terrestrial Ecosystems consider the following statements-

- 1. Nocturnal Behaviour of animals in desert ecosystems is an adaptation behaviour.
- 2. Predators are the primary consumers in the Terrestrial Ecosystems.
- 3. Terrestrial Ecosystems act as a Natural Carbon Sink of the Earth.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 only
- c) 1 and 2 only
- d) 2 and 3 only

Q.66) In the context of Terrestrial ecosystems, consider the following statements-

- 1. Wetlands are ecotones between Terrestrial and aquatic ecosystems.
- 2. The presence of diverse coniferous tree species are characteristics of Tropical Forests.
- 3. Dense canopies of vegetation are found in the Tundra Forest Ecosystem.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only



- b) 2 and 3 only
- c) 1 only
- d) 3 only

Q.67) Consider the following pairs-

Ecosystem components	and	its Characteristics/Examples
1. Tundra Ecosystem		Mosses, Lichens, Sledges
2. Lotic Ecosystem		Ponds, Marshes, Lakes.
3. Algae		Decomposers of the Aquatic ecosystem.

Select the correct answer using the code below-

- a) 1 and 3 only
- b) 2 and 3 only
- c) 1 only
- d) All of the above.

Q.68) Consider the following statements:

1. Desert ecosystem has the highest average net primary productivity.
2. Changes in the temperature profile with depth in an aquatic system is called Thermal Stratification.
3. Estuaries have a higher productivity as compared to the Grassland Ecosystems.

Which of the statements given above is/are incorrect?

- a) 2 and 3 only
- b) 1 only
- c) 1 and 3 only
- d) All of the above

Q.69) With reference to Ecosystems in general, consider the following statements:

1. Greater biodiversity leads to greater stability in Ecosystems.
2. All Ecotones are rich in biodiversity.
3. The Abiotic factors affecting the Aquatic ecosystems are Sunlight, water currents, Salinity, Microbes, Depth and Nutrients.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 and 3 only
- c) 1 and 2 only
- d) 3 only

Q.70) In context to Biological oxygen demand(BOD) consider the following statements:

1. Biological oxygen demand BOD is standard criterion for pollution assay in aquatic ecosystems.
2. Clean water has more biological oxygen demand than polluted water.

Which of the following statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) 1 and 2 both
- d) Neither 1 and 2

Q.71) With reference to the Aquatic Ecosystems, consider the following statements-

1. Blue carbon is the carbon stored in coastal and marine ecosystems whereas Green carbon is the carbon taken up by terrestrial ecosystems
2. Increased sediment deposition from the urban construction has resulted in greater turbidity and decreased photosynthesis in the aquatic ecosystem.
3. Only natural Wetlands have water purification functions.

Which of the statements given above is/are correct?

- a) 2 and 3 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) All of the above.

Q.72) In the context of Terrestrial and Aquatic Ecosystems, consider the following statements-

1. In the marine Ecosystems biomass pyramids are often inverted.
2. Pyramid of energy is inverted in Terrestrial Ecosystems.
3. Fungi are autotrophs which occupy the base of the terrestrial food web.

Which of the statements given above is/are correct?

- a) 2 and 3 only
- b) 1 and 3 only
- c) 1 only
- d) All of the above.

Q.73) With reference to Aquatic Ecosystems, consider the following statements-

1. Water depth affects the presence and the type of life in an aquatic ecosystem.



2. The rise in temperatures of the aquatic ecosystems will be less as compared to that of the terrestrial ecosystems.
3. Some marine organisms create bioluminescence as a form of an Adaptation.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above.

Q.74) Consider the following statements about the Desert Ecosystems of the Earth-

1. CAM Pathway in the plants helps to increase photorespiration and reduce water usage.
2. The lowest net primary productivity of the terrestrial environments is found in Deserts.
3. Xerophyte plants have fewer stomata to reduce water loss via transpiration.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 3 only

Q.75) Consider the following statements about the Terrestrial Ecosystems-

1. Fires act as a limiting factor in the Grassland Ecosystems.
2. Buttress roots in the Tundra Vegetation helps in the nutrient uptake.
3. The Rate of decomposition is high in the Tropical Rainforests.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 3 only

Q.76) With reference to Ecosystems, consider the following statements-

1. All Terrestrial animals are warm blooded in nature.
2. Plants in the aquatic ecosystem have well developed root systems.
3. Only Terrestrial animals breathe through their lungs.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 2 only
- c) 2 and 3 only

- d) All of the above.

Q.77) Consider the following statements-

1. An Ecotone is a population that is adapted to local environmental conditions.
2. The density of organisms present is lesser in the ecotone compared to the surrounding environments.
3. Areas experiencing edge effects might have higher predation levels than interior habitats.

Which of the statements given above is/are correct?

- a) 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 3 only

Q.78) In the context of the Terrestrial and Aquatic Ecosystems, consider the following-

1. Transpiration occurs in both Terrestrial and Aquatic Plants.
2. Inter-tidal Zones do not experience any Edge effects.
3. Podzols are the typical soils found in the boreal ecosystems.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 and 3 only
- c) 1 only
- d) All of the above.

Q.79) Consider the following statements-

1. Vivipary is a form of reproductive adaptation shown by Mangroves.
2. Pelagic Zones are areas that are attached to the shores of the surrounding lands.
3. Thermal pollution leads to an increase in the dissolved oxygen level in aquatic ecosystems.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 and 3 only
- c) 1 only
- d) All of the above.

Q.80) Consider the following statements about the Tidal Forest Ecosystems-

1. Tidal forests in India are found in the regions of Godavari, Krishna, and Mahanadi.
2. Destruction of mangroves is causing the decrease of the Kharai Camels.
3. Mangrove trees take up oxygen through their roots.



Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) None of the above

Q.81) Recently in the news, the 'Food Waste Index Report' was released by which of the following organizations ?

- a) Food and Agriculture Organization.
- b) United Nations Environment Program
- c) United Nations Children Fund
- d) World Food Program.

Q.82) The Greater One-Horned Rhinoceros is found in which of the following states in India.

- 1. Uttarakhand
- 2. Uttar Pradesh
- 3. West Bengal
- 4. Assam
- 5. Manipur

Select the correct answer from the codes given below.

- a) 1, 2, 3 and 4 only
- b) 2, 3 and 4 only
- c) 3 and 4 only
- d) 1, 2, 3, 4 and 5

Q.83) Which of the following statements is incorrect with respect to 'SVAMITVA SCHEME' ?

- a) It is a Central Sector Scheme that will pave the way for using property as a financial asset by villagers.
- b) The Scheme would be implemented throughout all the villages of the country in a phased manner.
- c) The Ministry of Panchayati Raj (MoPR) is the nodal ministry for implementing SVAMITVA scheme
- d) Legal ownership rights (Property cards/Title deeds) are issued by mapping of land parcels using drone technology.

Q.84) With reference to the National Security Act, consider the following statements:

- 1. A person detained in one state under the NSA cannot be shifted to another state for detention.
- 2. The Commissioner of Police can also exercise the provisions of the NSA.

Which of the statements given above is/are correct ?

- a) 1 only
- b) 2 only

- c) Both 1 and 2
- d) Neither 1 nor 2

Q.85) Which one of the following is not one of the dimensions of 'Vision 2030' released by the Government of India ?

- a) Through our space programme – Gaganyaan, India becoming the launch-pad of satellites for the World
- b) To make India pollution free by focusing on Electrical Vehicles and renewables.
- c) Self sufficiency in food production and producing food in the most organic way.
- d) To achieve 100% literacy (Both primary and secondary education) by 2030.

Q.86) With reference to the NISHTHA capacity building programme for Eklaya Model Residential Schools, consider the following statements

- 1. The States of Madhya Pradesh, Chhattisgarh and Himachal Pradesh comprised the first batch of states to complete the programme.
- 2. NITI Aayog collaborated with the Ministry of Education for implementation of this programme.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.87) Consider the following statements with reference to 'Renewable Energy Country Attractive Index'.

- 1. India's position deteriorated in the index as per the recently released Index.
- 2. The Index is published by the United Nations Environment Programme.
- 3. The United States of America tops the index followed by China and India.

Which of the statements is/are correct ?

- a) 1 and 2 only
- b) 2 only
- c) 3 only
- d) 1, 2 and 3

Q.88) The members of the National Board for Wildlife consists of which of the following ?

- 1. The Prime Minister
- 2. The Secretary to the Government of India, Ministry of Tribal Welfare



3. The Director, Botanical Survey of India
 4. The Chief of the Army Staff
 5. The Director, National Institute of Oceanography
- Select the correct answer from the codes given below:**

- a) 1, 2 and 3 only
- b) 2 and 5 only
- c) 1 and 5 only
- d) 1, 2, 3, 4 and 5

Q.89) The Union Ministry of Environment Forest and Climate Change (MoEFCC) in collaboration with World Wide Fund for Nature-India (WWF-India) and the Centre-funded Wildlife Institute of India (WII) plans to create for the first time the DNA profiles of which of the following species ?

- a) Tiger
- b) Greater One Horned Rhinoceros
- c) Indian Wild Dog
- d) Snow leopard

Q.90) 'Future for All- The need for human- wildlife coexistence' Report 2021, is published by

- a) International Union for Conservation of Nature
- b) United Nations Development Programme and World Wide Fund for Nature.
- c) United Nations Development Program and Greenpeace International.
- d) United Nations Framework Convention on Climate Change.

Q.91) Which of the following pairs are correctly matched ?

Protected Area. State/Union Territories.

1. Haliday Islands WLS - Andaman & Nicobar Islands.
2. Jorepokhri Salamander WLS - West Bengal.
3. Mahananda WLS - Odisha
4. Bori WLS - Maharashtra
5. Gandhi Sagar WLS - Madhya Pradesh.

Select the correct answer from the codes given below:

- a) 1, 2, 3 and 5 only
- b) 2 and 5 only
- c) 2, 3 and 4 only
- d) 1, 2, 3, 4 and 5

Q.92) Consider the following statements with respect to Project RE- HAB.

1. It is an initiative of the Ministry of Environment Forest and Climate Change in collaboration with Wildlife Institute of India.
2. The first phase of this project is implemented in the flood prone areas of Assam.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.93) Which of the following statements is incorrect with respect to the 'State Rooftop Solar Attractiveness Index-SARAL' ?

- a) SARAL is the first of its kind index to provide a comprehensive overview of state-level measures adopted to facilitate rooftop solar deployment.
- b) The index is released by the NITI Aayog.
- c) Business development and Investment climate are its major key aspects of SARAL.
- d) The index has ranked Karnataka at first followed by Telangana, Gujarat and Andhra Pradesh

Q.94) Consider the following statements with respect to the 'AIM- PRIME' (Atal Innovation Mission- Program for Researchers on Innovations, Market-Readiness & Entrepreneurship) program.

1. It is an initiative to promote and support science-based deep-tech startups & ventures across India.
2. AIM in collaboration with Bill & Melinda Gates Foundation (BMGF) to launch this nationwide program.
3. The benefits of this program are aimed at addressing specific issues through training and guidance over a period of 12 months.

Which of the above statements given is/are correct ?

- a) 1 and 2 only
- b) 1 only
- c) 2 and 3 only
- d) 1, 2 and 3.

Q.95) Consider the following statements with reference to 'Pinaka Rocket System':

1. It is developed by the Indian Space Research Organization in collaboration with the Polish Space Agency.
2. Pinaka Rocket System is the upgraded version of Ramjet Rocket Propulsion System with greater average cruise speed.

Which of the above statements is/are correct ?

- a) 1 only



- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.96) Recently in the news, 'Ambition to Impact: Opportunities for Global Collaboration in India's Clean Energy Economy' is a joint collaboration of ?

- a) NITI Aayog and World Bank.
- b) Ministry of Environment Forest and Climate Change and Food and Agriculture Organization.
- c) Ministry of New and Renewable Energy, the Permanent Mission of India to the UN at New York, and the Council on Energy, Environment and Water.
- d) International Solar Alliance and United Nations Framework for Climate Change.

Q.97) 'The India Story' recently in the news is related to ?

- a) Measures taken by the Union Ministry of Health and Family Welfare to fight with the ongoing covid-19 pandemic.
- b) Mitigation measures taken by the Ministry of Environment Forest and Climate Change to reverse the effect of climate change.
- c) ISRO's publication highlighting the success stories with respect to space programs in the last decade.
- d) Ministry of New and Renewable Energy compilation of Indian initiatives that are shaping India's energy transition.

Q.98) Consider the following statements:

- 1. One Nation One Standard mission was launched by the Bureau of Indian Standards (BIS) to ensure quality products in the country
- 2. RDSO (Research Design & Standards Organization) becomes the FIRST Institution to be declared SDO under one "One Nation One Standard" mission on BIS (Bureau of Indian Standards)

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.99) Recently, India signed a Memorandum of Understanding with which of the following countries for extraction, mining and beneficiation of lithium ?

- a) Chile
- b) Argentina
- c) Australia

- d) Brazil

Q.100) Which of the following are among the most common applications of Lithium-ion batteries?

- 1. Solar power storage
- 2. Batteries of Mobiles, laptops and similar consumer electronic goods
- 3. Electric mobility
- 4. Smartwatches that can track sleep and monitor pulse rate.

Select the correct answer using the codes given below

- a) 1 and 3 only
- b) 2 and 3 only
- c) 1, 3 and 4 only
- d) 1, 2, 3 and 4

Q.1) Ans: A

Exp:

- **Statement 1 is Correct : Ecological Pyramid :** The steps of trophic levels expressed in a diagrammatic way are referred to as ecological pyramids. The **food producer forms the base of the pyramid and the top carnivore forms the tip.** Other consumer trophic levels are in between.
- The pyramid consists of a number of horizontal bars depicting specific trophic levels which are arranged sequentially from primary producer level through herbivore, carnivore onwards. The length of each bar represents the total number of individuals at each trophic level in an ecosystem.
- The ecological pyramids are of three categories: Pyramid of numbers, Pyramid of biomass and Pyramid of energy or productivity.
- **Statement 2 is Incorrect :** The number, biomass and energy of organisms may gradually decrease or increase with each step from the producer level to the consumer level and the diagrammatic representation assumes a pyramid shape. The decrease or increase depends on the ecosystem taken into consideration.
 - For example - In the Grassland Ecosystem, the Pyramid of number is upright while that in aquatic systems is Inverted. It can be understood in following diagram :

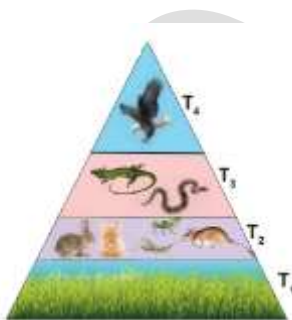
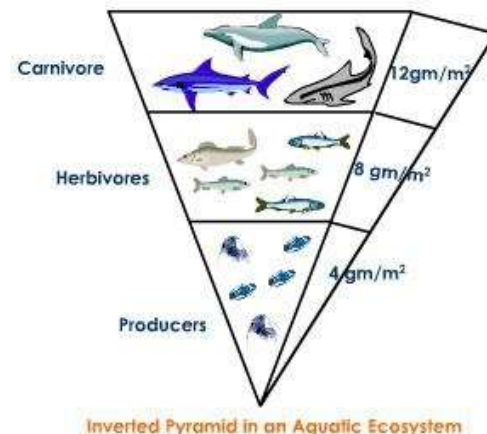


Figure 12 a: Pyramid of numbers in grassland ecosystem



Q.2) Ans: B

Exp:

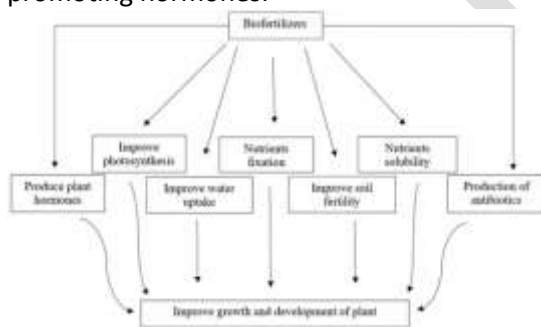
- **Pyramid of numbers** shows the number of the producers, herbivores and the carnivores at their successive Trophic levels. This pyramid can be either upright, or inverted or partially upright
 - **Option a is Correct : In a forest ecosystem -** the producers are large sized trees which make the base of the Pyramid. The herbivores such as fruit eating birds, deer, elephants etc. make the primary consumers and are less than primary producers. After that, the number goes down at each successive level. Thus the pyramid of numbers is **upright** in the forest ecosystem.
 - **Option b is Incorrect : In a grassland ecosystem -** the number of producers (mainly grasses) is always maximum, followed by decreasing numbers at second Trophic level (herbivores), third Trophic level (carnivores) and least number of apex predators. Thus, a pyramid of numbers is **upright** in the grassland ecosystem.
- **Pyramid of Biomass** shows the biomass of the producers, herbivores and the carnivores at their successive Trophic levels.
 - **Option d is Correct : In Terrestrial Ecosystem -** (Grasslands, forests) -: The biomass generally decreases at each higher Trophic level from plants via herbivores to carnivores. Thus the Pyramid of biomass is **upright** in the grassland ecosystem.
 - **Option c is Correct : In the aquatic system -** the biomass can increase at higher Trophic levels. For example, in Oceans, the food

chain typically starts with phytoplankton and ends at predatory fish, which has the largest biomass. Thus, the pyramid of biomass is **inverted** in the aquatic systems including marine, ponds etc.

Q.3) Ans: A

Exp:

- **Statement 1 is Correct** : Biofertilizers are a substance which contains living microorganisms which, when applied to seed, plant surfaces, or soil, promote the growth by increasing the supply or availability of primary nutrients to the plant.
- **Statement 2 is Correct** : Biofertilizers add nutrients through the natural processes of fixing atmospheric nitrogen, solubilizing Phosphorus, and stimulating plant growth through the synthesis of growth promoting substances.
- **Statement 3 is Incorrect** : Biofertilizers harness atmospheric nitrogen and make it available directly to the plants. It increases phosphorus uptake by solubilising and releasing unavailable phosphorus. It also enhances root proliferation due to release of growth promoting hormones.



Q.4) Ans: C

Exp:

- **Statement 1 is Correct** : Estuaries are located where rivers meet the sea. Estuaries are water bodies where the flow of freshwater from rivers mixes with salt water transported by tide, from the ocean
 - An estuary is a semi enclosed coastal body of water with one or more rivers

flowing into it. It has a free connection with the open sea.

- Estuaries are usually biologically highly productive zones. Many estuaries are locations of some of the major seaports. Most of India's major estuaries occur on the east coast, in contrast, the estuaries on the west coast are smaller.

- **Statement 2 is Incorrect** : Estuaries are homes to plants & animals which have high salt tolerance since the mixing of sea water increases the salt content more than normal in water of estuaries. To survive in this condition, the plants, trees etc which have high salt tolerance adapt to the ecosystem.
- **Statement 3 is Correct** : Estuaries create a natural barrier that absorbs the energy of the waves and prevents them from flooding cities.

Q.5) Ans: A

Exp:

- **Statement 1 is Correct** : The **Ramsar Convention on Wetlands** is the intergovernmental treaty that provides the framework for the conservation and wise use of wetlands and their resources.
 - The Convention was adopted in the Iranian city of Ramsar in 1971 and came into force in 1975.
 - Since then, almost 90% of UN member states, from all the world's geographic regions, have acceded to become "Contracting Parties".
 - Every three years, representatives of the contracting parties meet as the Conference of the Contracting Parties (COP), the policy-making organ of the convention which adopts decisions (resolutions and recommendations) to administer the work of the convention and improve the way in which the parties are able to implement its objectives.
- **Statement 2 is Incorrect** : The Ramsar Convention works closely with six other organisations known as international organization partners (IOPs). These are -: **Birdlife International; IUCN; International Water Management Institute (IWMI); Wetlands International; WWF International; Wildfowl & Wetlands Trust (WWT)**.



- IUCN provides administrative services to the Convention.

Q.6) Ans: C

Exp:

- **Statement 1 is Correct** - The adaptation is any attribute of the organism (**morphological, physiological, and behavioural**) that enables the organism to survive and reproduce in its habitat. Many adaptations have evolved over a long evolutionary time and are genetically fixed. Some of the examples are as follows:
 - **Morphological:** Many desert plants have a thick cuticle on their leaf surfaces and have their stomata arranged in deep pits (sunken) to minimise water loss through transpiration.
 - **Physiological** : Many tribes living in the high altitude of Himalayas have a higher red blood cell count.
 - **Behavioural:** Animals migrating temporarily to a less stressful habitat.
- **Statement 2 is Correct** : According to Charles Darwin's theory of evolution by natural selection, the organisms adapt to their environment so that they could persist and pass their genes onto the next generation.
 - **Adaptation differs from Acclimatization.** Both terms are about changes. However, acclimatization is the physiological adjustment to the new conditions but it does not entail increasing species diversity as adaptation does. For a trait to be considered as an adaptation, it has to be heritable, functional, and increases fitness.

Q.7) Ans: B

Exp:

- **Statement 1 is Correct - Ecological niche** is a term for the position of a species within an ecosystem, describing both the range of conditions necessary for persistence of the species, and its ecological role in the ecosystem.
 - Ecological niche subsumes all of the interactions between a species and

the biotic and abiotic environment, and thus represents a very basic and fundamental ecological concept.

- **Statement 2 is Incorrect** : Niche Diversity and Stability has a Direct Relation. The greater the niche diversity, the more is ecosystem stability because of the larger number of pathways for the flow of energy and less fluctuation of the species population.
- **Statement 3 is Correct** : Dominant species occupy an extensive and broader ecological niche in comparison to less dominant species.

Q.8) Ans: A

Exp:

Adaptations found in mangroves are:

- **Statement 1 is correct:** Mangroves exhibit a **Viviparity mode of reproduction**. i.e. **seeds germinate in the tree itself** (before falling to the ground). This is an adaptive mechanism to overcome the **problem of germination** in saline water.
- **Statement 2 is correct:** **Adventitious roots** which emerged from the main trunk of a tree above ground level are called **stilt roots**.
- **Statement 3 is correct:** They produce **pneumatophores** (air roots up from the mud) to overcome the **respiration problems** in the anaerobic soil conditions.
- **Statement 4 is incorrect:** **Leaves are thick** and contain **salt secreting glands**.
- **Statement 5 is correct:** Most mangrove vegetation has **lenticellated bark** which facilitates **more water loss, producing coppices**.

Extra Edge by OnlyIAS

Other characteristics of Mangroves:

- **Adaptation to salinity** and to water-logged anaerobic soil is **high**
- Require **high solar radiation** and have the ability to **absorb fresh water from saline water**

Q.9) Ans: B

Exp:

Criteria for identification of wetlands of national importance under Ramsar Convention:



- If a site regularly supports **20,000 or more water birds**.
- If a site regularly supports **1% of the individuals** in a population of **one species or subspecies** of waterbirds.
- If a site supports a significant proportion of **indigenous fish subspecies, species or families**, life-history stages, species interactions and/or populations that are **representative of wetland benefits** and/or values and thereby contributes to global biological diversity.
- If a site supports **vulnerable, endangered, or critically endangered species**; or threatened ecological communities.

Extra Edge by OnlyIAS

Other criteria for identification of wetlands of national importance:

- If a site contains a **representative, rare, or unique** example of a **natural or near-natural wetland type** found within the appropriate biogeographic region.
- If a site supports **populations of plant and/or animal** species important for **maintaining the biological diversity** of a particular biogeographic region.
- If a site supports **plant and/or animal species** at a **critical stage in their life cycles**, or provides **refuge during adverse conditions**.
- If a site is an important **source of food for fishes**, a spawning ground, nursery and/or migration path on which fish stocks, either within the wetland or elsewhere, depend.
- If a site is an **important source of food and water resources**, increased possibilities for **recreation and ecotourism**, improved scenic values, educational opportunities, conservation of cultural heritage (historic or religious sites).

region show decrease in forest cover. The loss in the North East is attributed primarily to the traditional farming practice of **Shifting Cultivation**.

- **Option B is incorrect:** The **total mangrove cover** in the country is **4,975 sq km**. An **increase of 54 sq Km** in mangrove cover has been observed as compared to the previous assessment of 2017. Top three states showing mangrove cover increase are **Gujarat (37 sq km)** followed by **Maharashtra (16 sq km)** and **Odisha (8 sq km)**.
- **Option C is correct:** The **total carbon stock** in the country's forest is estimated at **7,124.6 million tonnes** and there is an **increase of 42.6 million tonnes** in the country's carbon stock as compared to the last assessment of 2017.
- **Option D is incorrect:** Area-wise **Madhya Pradesh** has the **largest forest cover** in the country followed by **Arunachal Pradesh, Chhattisgarh, Odisha and Maharashtra**. In terms of **forest cover as percentage of total geographical area**, the top five States are **Mizoram (85.41%), Arunachal Pradesh (79.63%), Meghalaya (76.33%), Manipur (75.46%) and Nagaland (75.31%)**.

Reference:

<https://pib.gov.in/PressReleaseDetailm.aspx?PRID=1597987>

Q.11) Ans: A

Exp:

- **Statement 1 is correct:** Under the rules the Central government has **decentralized the powers** for management of wetlands by **empowering the states and union territories to identify and manage their wetlands**.
- **Statement 2 is incorrect:** The rules provide for the setting up of a **State Wetlands Authority** in each State and union territories that will be headed by the **State's environment minister** and include a range of government officials. They will also include one expert each in the fields of wetland ecology, hydrology, fisheries, landscape planning and socioeconomics to be nominated by the state government.
- The rules stipulate setting up of a **National Wetlands Committee**, which will be headed by the **MoEFCC Secretary**, to monitor implementation of these rules by the States.

Q.10) Ans: C

Exp:

- **Option A is incorrect:** According to the India State of Forest Report, 2019, the **total Forest and Tree cover** of the country is **8,07,276 sq km** which is **24.56%** of the geographical area of the country. There is a **decrease of forest cover** to the extent of **765 sq km (0.45 per cent)** in the northeast India region. **Except Assam and Tripura**, all states in the



- **Statement 3 is incorrect:** The rules are applicable to the:
 - Wetlands categorised as '**wetlands of international importance**' under the Ramsar Convention
 - Wetlands as **notified** by the **Central Government, State Government and Union Territory Administration**.

Reference:

<https://www.livemint.com/Politics/y6Tr3tkrr3q28AmGKaBFII/Environment-ministry-notifies-new-wetland-rules.html>

Q.12) Ans: C

Exp:

- **Statement 1 is correct:** Many plants in the desert **conserve water** because of **reduced size of leaves** or **not having any leaves at all**. For example: **cactus**. Many cacti have **spines in place of leaves**, which conduct **photosynthesis and catch dew** when the climate is right. These small structures **also reflect light**, further reducing water loss.
- **Statement 2 is correct:** In some plants, even the **stem contains chlorophyll for photosynthesis**. The stems also **store water**, so they appear pronounced and fleshy.
- **Statement 3 is incorrect:** These plants **spread their roots far and wide underground**, yet close enough to the surface to soak up any available water during the rare rainstorms.
- **Statement 4 is correct:** Most desert plants are **inactive for much of the year**. During **dry periods** they do **not perform** many water-draining functions such as **photosynthesis**.

Reference:

<https://sciencing.com/do-desert-plants-adapt-environment-6526946.html>

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Other adaptations:

- These plants are **mostly shrubs**.
- Leaves and stems are **succulent and water storing**.
- These plants **germinate, bloom and reproduce only during** the short **rainy season**, and not in summer and winter.

Q.13) Ans: A

Exp:

Cold water corals are found at a depth **up to 6,000 m** below the ocean's surface, where water is **icy cold** and the light dim or absent. They are also called **deep-sea corals**. deep-sea corals occur in the waters of the United Kingdom, Australia, New Zealand, Canada, Ecuador, Japan, Norway, and the United States.

- **Option A is Correct:** Unlike warm water corals, deep-sea corals **don't need sunlight**. They obtain the **energy and nutrients** they need to survive by **trapping tiny organisms** in passing currents.
- **Option B is Incorrect:** The **range of size** of deep-sea corals is **very high**. They are found as single polyps as **small as a grain of rice**, tree-like colonies that are **as tall as 10m**.
- **Option C is Incorrect:** The growth rate of deep-water corals is around 1-2.5 cm/year, whereas that of warm-water corals is 10-20 cm/year.

<https://ocean.si.edu/ecosystems/coral-reefs/deep-sea-corals>

<https://web.archive.org/web/20100221152237/http://coris.noaa.gov/about/deep/>

Q.14) Ans: C

Exp:

- **Statement 1 is correct:** Grasslands occupy **nearly 24 percent of the geographical area** in India and are found where rainfall is **about 25-75 cm per year**, not enough to support a forest, but more than that of a true desert.
- **Statement 2 is incorrect:** **Steppe formations** occupy large areas of **sandy and saline soil**; in **western Rajasthan**, where the climate is **semi-arid**, **average rainfall is less than 20 cm** a year with a **dry season of 10 to 11 months**, and a **large variation in rainfall**.
In the **central and eastern parts of Rajasthan**, where the **rainfall is about 50 cm per year** and the **dry season is of six to eight months**, dry **savanna grazing ecosystems** have developed.
- **Statement 3 is correct:** **Saurashtra region** in Gujarat is characterized by **hot and semi-arid climate** and typically **thorn scrub vegetation** with a high proportion of graminoids commonly referred to as **scrub savanna**. These areas are **locally called 'Vidi'** and have been used for livestock grazing by the pastoral communities.



Reference:

<https://www.indiawaterportal.org/faqs/grasslands>
https://www.researchgate.net/publication/303686549_The_Vidis_of_Saurashtra

Q.15) Ans: D

Exp:

- **Statement 1 and 2 are correct:** Kelp Forests are **underwater ecosystems** formed in **shallow water** by the dense growth of several different species known as kelps. Though they look very much like plants, kelps are **actually extremely large brown algae**. Kelp thrives in **cold, nutrient-rich waters**. Because kelp attaches to the seafloor and eventually grows to the water's surface and relies on **sunlight** to generate **food and energy**, kelp forests are **always coastal** and require **shallow, relatively clear water**.
- **Statement 3 is correct:** They are known for their **high growth rate**. Some varieties grow **as fast as half a meter a day**, ultimately reaching **30 to 80 meters**. Kelp forests are recognized as one of the most **productive and dynamic ecosystems** on Earth. They can disappear and reappear based on the oceanographic conditions and the population sizes of their primary herbivores.

Reference:

<https://oceana.org/marine-life/marine-science-and-ecosystems/kelp-forest>

Q.16) Ans: D

Exp:

- **Statement 1 is correct:** Eutrophication is a **syndrome of ecosystem** response to the **addition of artificial or natural nutrients** such as **nitrates** and **phosphates** through **fertilizer, sewage**, etc. that fertilize the aquatic ecosystem. It is primarily caused by the **leaching of phosphate** and/or **nitrate** containing fertilisers from **agricultural lands into lakes or rivers**.
- **Statement 2 is correct:** Reduction in sunlight for the green plants in the lakes/rivers **reduces the oxygen level** in the lakes or rivers. Further **oxygen is taken up by microorganisms** which feed off the dead algae during the decomposition process. Due to **reduced oxygen levels**, fishes and other

aquatic organisms suffocate and they die. These **anaerobic conditions** can promote **growth of bacteria** which produces **toxins deadly to aquatic organisms**, birds and mammals.

- **Statement 3 is correct:** The growth of **green algae** which we see in the lake surface layer is the **physical identification of Eutrophication**. Some algae and blue-green bacteria thrive on the excess ions and a population explosion covering almost the entire surface layer is known as **algal bloom**.

Reference:

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Q.17) Ans: C

Exp:

The above passage describes **Tropical wet evergreen forests**.

- **Tropical dry evergreen forests:** These are found along **Tamil Nadu and Andhra Pradesh coast**. It is mainly **hard-leaved evergreen trees** with fragrant flowers, along with a few deciduous trees.
- **Tropical dry deciduous forests:** Dry deciduous forests are found **throughout the northern part** of the country **except in the North-East**. It is also found in Madhya Pradesh, Gujarat, Andhra Pradesh, Karnataka, and Tamil Nadu. The **canopy** of the trees does **not normally exceed 25 metres**. The common trees are the **sal**, a variety of **acacia**, and **bamboo**.
- **Tropical moist deciduous forests:** Moist deciduous forests are found **throughout India except in the western and the north-western regions**. The trees are **tall**, have **broad trunks**, **branching trunks** and **roots** to hold them firmly to the ground. Some of the **taller trees shed their leaves** in the **dry season**. These forests are dominated by **sal** and **teak**, along with **mango**, **bamboo**, and **rosewood**.

Q.18) Ans: A

Exp:

- **Statement 1 is incorrect:** Swamps are **forested wetlands** that occur around the world in **riverine**, **lacustrine**, and **palustrine habitats**. They are different from marshes in a sense that **swamps**



are **dominated by trees**, whereas in **marshes** the plant life mainly **consists of grasses**.

- **Statements 2 and 3 are correct:** Swamps are characterized by **mineral soils** and exist in areas with **poor drainage and sufficient water supply** to keep the ground waterlogged, and they have a **high enough supply of minerals** in the water to stimulate **decay of organisms** and **prevent** the accumulation of **organic materials**.

Reference:

<https://www.britannica.com/science/swamp>

<https://www.sciencedirect.com/topics/agricultural-and-biological-sciences/swamps>

Q.19) Ans: A

Exp:

- **Statement 1 is correct:** An **estuary** is an area where a **freshwater river or stream meets the ocean**. The **complete salinity** ranges from **0-35 ppt** and is seen from the head (river end) to the mouth (sea end) of an estuary. In estuaries, **water level and salinity** rise and fall with the **tides** and **seasons**. During the **rainy season**, rivers may flood the estuary with **freshwater**. During the **dry season**, the **outflow from rivers** may slow to a trickle. The estuary shrinks, and becomes **much more saline**.
- **Statement 2 is incorrect:** When **dense, salty seawater flows** into an estuary, it has an **estuarine current**. **High tides** can create **estuarine currents**. Saltwater is heavier than freshwater, so **estuarine currents sink** and move **near the bottom of the estuary**. When **less-dense freshwater** from a river flows into the estuary, it has an **anti-estuarine current**. Anti-estuarine currents are strongest near the **surface of the water**. Heated by the sun, **anti-estuarine currents** are much **warmer** than estuarine currents.

Reference:

<https://www.nationalgeographic.org/encyclopedia/estuary/>

Q.20) Ans: A

Exp:

- **Pair 1 is correct:** **Littoral and swamp forests** are found along the **Andaman and Nicobar Islands** and the **delta** area of the **Ganga and the Brahmaputra**. They have roots that consist of **soft tissue** so that the plant can **breathe in the water**.
- **Pair 2 is correct:** **Tropical thorn forests** occur in the areas which receive **rainfall less than 50 cm**. These consist of a variety of **grasses and shrubs**. It includes semi-arid areas of **southwest Punjab, Haryana, Rajasthan, Gujarat, Madhya Pradesh and Uttar Pradesh**.
- **Pair 3 is correct:** **Sub alpine forests** extend from **Kashmir to Arunachal Pradesh** between **2900 to 3500 metres**. In the **Western Himalayas**, the vegetation consists mainly of **juniper, rhododendron, willow, and black currant**. In the **eastern parts**, **red fir, black juniper, birch, and larch** are the common trees. Due to heavy rainfall and high humidity the timberline in this part is higher than that in the West.
- **Pair 4 is incorrect:** In the **North**, **Montane wet temperate forests** are found in the region to the **east of Nepal into Arunachal Pradesh**, receiving a **minimum rainfall of 2000 mm**. In the **South**, it is found in parts of the **Nilgiri Hills**, the higher reaches of **Kerala**. The forests in the northern region are denser than in the South. **Rhododendrons** and a variety of ground flora can be found here.

Reference:

Geography NCERT Class 11

Q.21) Ans: C

Exp:

- **Pair 1 is correct:** **Kolleru Lake** is the **largest shallow freshwater lake in Asia**. Kolleru is located between **Krishna and Godavari deltas** spanning over two districts – **Krishna and West Godavari**. It is a **hot spot for migratory** as well as resident birds. **Local vaddi community** is dependent on this lake for fishing.
- **Pair 2 is incorrect:** **Ashtamudi lake** is a natural backwater in the **Kollam district in Kerala**. River **Kallada** (forms Munroe island) and **Pallichal** drains into it. The **National Waterways-3** passes through it. It is an **extensive estuarine system**, the second largest in Kerala State. The site supports a number of **mangrove species**.

- **Pair 3 is correct:** Lonar lake is an endorheic (closed basin) crater lake formed by a meteorite impact. It is high in salinity and alkalinity due to the lack of an outflow. Recently, the colour of Lonar lake water had turned pink due to a large presence of the salt-loving "haloarchaea" microbes.
- **Pair 4 is correct:** The Kabartaal Wetland in Bihar is an important stopover along the Central Asian Flyway, with 58 migratory waterbirds using it to rest and refuel. Five critically endangered species inhabit the site, including three vultures – the red-headed vulture, white-rumped vulture and Indian vulture – and two waterbirds, the sociable lapwing and Baer's pochard.

Reference:

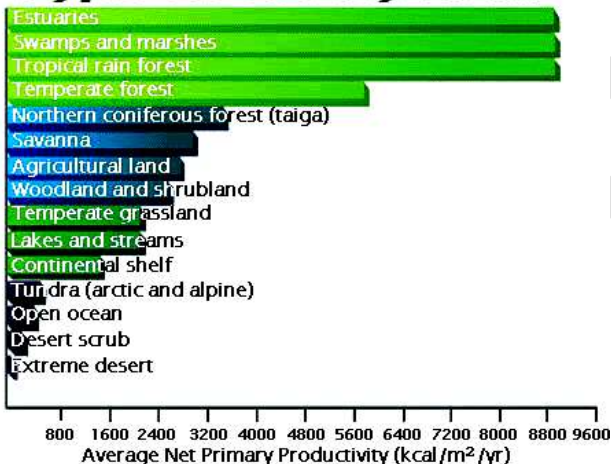
<https://rsis.ramsar.org/ris/2436>

<https://www.pmfias.com/ramsar-sites-india-ramsar-convention-wetlands/>

Q.22) Ans: C

Exp:

Type of Ecosystem



- Primary production is defined as the amount of biomass or organic matter produced per unit by the plants during photosynthesis and the rate of biomass production by the plants is called primary productivity.
- Gross Primary Productivity (GPP) of an ecosystem is the weight of production of organic matter during photosynthesis. But a considerable amount of GPP is utilised by the plants in respiration.

GPP — Respiration losses = Net Primary Productivity

- It is the Net Primary Productivity (NPP) which is the available biomass for consumption by the heterotrophs.

Reference:

<https://globalchange.umich.edu/globalchange1/current/lectures/klings/energyflow/energyflow.html>

Q.23) Ans: D

Exp:

- **Statement 1 is correct:** Cold desert/temperate desert of India includes areas of Ladakh, Leh, Kargil, Spiti valley of Himachal Pradesh and some parts of Uttarakhand and Sikkim.
- **Statement 2 is correct:** These arid areas are not affected by the Indian monsoons because they lie in the rain-shadow of the Himalayan mountain systems.
- **Statement 3 is correct:** The soil of cold deserts is sandy to sandy loam. It is neutral to slightly alkaline, has poor organic matter content and has low water retention capacity.

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Other characteristics of the Cold Desert:

- These are characterised by extreme cold weather and denuded terrain that are not suitable for plant growth. Isolated, scattered and over-grazed herbaceous shrubs are found. Grazing period is less than 3-4 months.
- The conditions are arid and the temperature is less than 0°C for most of the period, dropping to -50°C during winter.
- Mean annual rainfall is less than 400mm and heavy snowfall occurs between November and March.
- Wind erosion is more common.

Q.24) Ans: B

Exp:

- **Statement 1 is incorrect:** Desertification refers to destruction of biological potential of land which

can ultimately lead to **desert-like conditions in arid and semi-arid areas**. In India, desertification is not confined to Rajasthan, Gujarat, Punjab and Haryana. For example, in the **northeast region** due to **jhum cultivation**. According to the **Desertification and Land Degradation Atlas, 2016**, **29.32%** of the Total Geographic Area of the country is going through the **process of land degradation**. Around **24%** of this area is contributed by **Rajasthan, Maharashtra, Gujarat, Jammu & Kashmir, Karnataka, Jharkhand, Odisha, Madhya Pradesh and Telangana**.

- **Statement 2 is correct:** The **United Nations Convention to Combat Desertification** is the **sole legally binding** international agreement linking environment and development to sustainable land management. It was adopted in **1994**. The Convention addresses specifically the **arid, semi-arid, and dry sub-humid areas**, known as the drylands, where some of the most vulnerable ecosystems and peoples can be found.
- **Statement 3 is incorrect:** **India** hosted the **14th session of Conference of Parties (COP 14)** of United Nations Convention to Combat Desertification (**UNCCD**) in September 2019 and is striving towards achieving the national commitments of **Land Degradation Neutrality (LDN)** and restoration of **26 Million ha of degraded land by 2030**.

Reference:

<https://pib.gov.in/PressReleaseFramePage.aspx?PRID=1727987>

<https://www.downtoearth.org.in/news/agriculture/desertification-in-india-slash-and-burn-farming-destroys-nagaland-66600>

<https://www.isro.gov.in/desertification-and-land-degradation-atlas-released>

Q.25) Ans: B

Exp:

- **Organic wastes** are domestic wastes, dead and decaying plants and animals, animal wastes and excreta, etc.
 - **Phenol** is an organic pollutant released mainly from the Iron and Steel industry. Phenol also known as carbolic acid is made of carbon, hydrogen, and oxygen.
 - **Glycerol** is an organic pollutant released from the soap and detergent industry.

- **Proteins and carbohydrates** are the organic pollutants from the pharmaceutical industry.

- **Inorganic pollutants** are in the form of chemical substances which are dangerous for the humans as they cannot be excreted by the human body. The ingested food is converted into carbohydrates which is a source of energy stored in the form of glucose.

- Silica and mercury are the inorganic pollutants released into the atmosphere.

Q.26) Ans: C

Exp:

- **Statement 1 is incorrect:** Presence of organic and inorganic wastes in the water decreases the Dissolved Oxygen (DO) of the water as much of the oxygen is used by the decomposers in the water body to carry out the decomposition of these wastes. Oxygen is required by aquatic species for survival. Hence, the presence of organic matter is an important indicator of quality of the aquatic ecosystem.
- **Statement 2 is correct:** **Biological Oxygen Demand** refers to the amount of oxygen needed by the bacteria to decompose the organic matter present in the water. **Chemical oxygen demand** refers to the oxygen required for the decomposition of both organic and inorganic matter present in the water. The Greater the Chemical Oxygen Demand less is the amount of Dissolved Oxygen and the availability of the oxygen for respiration is reduced.
- **Statement 3 is incorrect:** A number of factors determine the presence of Dissolved Oxygen in the water body. It includes photosynthesis, respiration by organisms, decomposition of organic wastes, movement of surface water, etc.

Q.27) Ans: B

Exp:

- **Stubble burning** refers to the burning of the crop residue after the harvesting of wheat and paddy crops. It prepares the field for the next season. Stubble burning is one of the major reasons for pollution in and around Delhi during the winter months.



- A number of factors are responsible for the burning of stubble in the fields which includes **use of combine harvesters** that leave behind small residues in the fields. These residues are difficult to collect and are hence usually burnt. **Intensification of the crop** requires the ground to be immediately cleared for the next crop and hence burning the residue is the easiest option.
- **Paddy transplantation** is not itself the reason for the burning of the stubble. **Cultivation of short duration rice varieties** would provide an increased window for the clearing of the fields and **crop diversification** would be another method to control stubble burning.
- **The Happy Seeder Machine** is a tractor operated machine used for direct sowing of wheat and hence prevents stubble burning. It also increases the productivity of the crop.

Source:

<https://www.downtoearth.org.in/blog/agriculture/stubble-burning-a-problem-for-the-environment-agriculture-and-humans-64912>

Q.28) Ans: D

Exp:

- **Statement 1 is incorrect:** **Bioremediation** involves the degradation of environmental contaminants into lesser toxic forms with the use of microorganisms like Bacteria and fungi. It can be done either **ex-situ** or **in-situ**. **In-situ bioremediation** involves the degradation by microbes in the site of the contamination while the **ex-situ** involves its degradation outside the site of contamination. A number of techniques are used to bring about ex-situ and in-situ bioremediation. The in-situ method comprises techniques like Bioventing, Bioaugmentation, and Biosparging. In case of Bioaugmentation, microorganisms are imported to the contaminated site to speed up the process of degradation. Hence, it is an in-situ technique.
- **Statement 2 is incorrect:** Bioremediation is a cost effective and environment friendly technique for bringing about degradation of the pollutants in the environment. It can be used to target a particular pollutant specifically. However, the

technique works only in cases of those compounds that are degradable. It does not apply to the compounds that are not degradable.

Q.29) Ans: B

Exp:

- **Statement 1 is correct:** In the pursuance of the Supreme Court directive, the **Environment Pollution (Prevention and Control) Authority** was created by the Ministry of Environment, Forests, and Climate Change under the provisions of the **Environment (Protection) Act, 1986** to prevent and control the pollution of environment in the National capital region. The authority has the power to take up cases related to the quality of the environment, environmental pollution, etc. suo moto or based on the complaints received by it.
- **Statement 2 is incorrect:** It deals with the implementation of the **Graded Response Action Plan** which deals with the set of steps to be taken when the level of air pollution passes a certain threshold. It classified the pollution into various categories like moderate to poor air, very poor, severely polluted, emergency, etc. The **National Clean Air Programme** on the other hand concentrates on the 112 non-attainment cities and focuses on the control of PM 2.5 and PM 10. It is implemented by the **Central Pollution Control Board**. It seeks to reduce the concentration of particulate matter by 20 to 30% by 2024 keeping 2017 as the base year.
- **Statement 3 is incorrect:** **Safe Water Release (SWAS), Safe Thermite Cracker (STAR), Safe Minimal Aluminum (SAFAL)** are the green crackers that are free from the chemicals like arsenic, barium, lead, etc. They are developed by **CSIR** and are not the initiative of the Environment Pollution (Prevention and Control) Authority.

Source: <https://epca.org.in/EPCA-reconstitution-order-dt.3Oct2018.pdf> ;

<https://indianexpress.com/article/explained/explained-what-is-grap-delhi-ncrs-action-plan-as-air-pollution-increases-6719746/> ;

<https://pib.gov.in/PressReleasePage.aspx?PRID=1551171> ;

<https://pib.gov.in/PressReleasePage.aspx?PRID=1655>



203#:~:text=The%20Central%20Government%20launched%20National,2017%20as%20the%20base%20year

Q.30) Ans: D

Exp:

- **Statement 1 is incorrect:** Certain areas are more prone to acid rain as compared to others. **Acid rains** are caused when the **oxides of Sulphur and Nitrogen** are mixed with the moisture in the atmosphere. It comes down to the surface of the earth as acid rain and causes huge damage to the aquatic and terrestrial plants along with damage to the man-made structures. They are mostly found in the industrialized areas and also in the mountainous regions which are subjected to huge rainfalls. They are also found in the mountainous areas with thin soils.
- **Statement 2 is incorrect:** **Rain water** is considered acidic if it has a pH value of less than 5.6. It is found in the atmosphere in the form of both wet and dry deposition. If the acids in the atmosphere fall in the form of rain, snow, fog, mist, etc. It is the wet deposition. Whereas if it blows from one place to another in a dry area, then it is called dry deposition.

Q.31) Ans: A

Exp:

- **Statement 1 is correct:** **Acid rain** causes heavy damage to the forests, lakes, rivers, deserts, and almost all the ecosystems along with its inhabitants. Similarly, lakes found can also have varied concentrations of acids in them. There can be both naturally and anthropogenically induced acidification of the lakes. In the case of naturally acidified lakes, the concentration of dissolved organic carbon is high. A naturally acidified lake appears discoloured due to the presence of humic substances in it. Anthropogenically acidified lakes on the other hand have low concentration of organic carbon.
- **Statement 2 is correct:** The naturally acidified lakes are not transparent rather they appear yellow or brownish due to the humic content present in them. The anthropogenically acidified lakes on the other hand appear transparent with very clear waters.

- **Statement 3 is incorrect:** While the naturally acidified lakes are well buffered, the anthropogenically acidified ones are poorly buffered with some species of aquatic life missing or in very low concentration. The naturally acidified lakes on the other hand are abundant with aquatic life.

Q.32) Ans: A

Exp:

- **Statement 1 is correct:** **Parthenocarpic fruit** is the one that matures without the formation of the seed. In cases when the seeds are formed without fertilization, it is called the agamospermy. It is also a kind of parthenogenesis. Bananas, cucumbers, grapes, etc. are examples of the parthenocarpic fruits.
- **Statement 2 is incorrect:** Plants that give beverage drinks are known as the beverage plants. It could be either alcoholic or non-alcoholic in nature. Alcoholic ones carry at least one or more hydroxyl groups whereas the non-alcoholic ones are alkaloids and contain caffeine.

Q.33) Ans: B

Exp:

- Acid rains are formed with the reaction of compounds containing Sulphur, nitrogen, and other acids with the moisture in the atmosphere. Other acids include Phosphoric acid, hydrochloric acid, carbonic acids, chlorine, etc. Hence, Sulphur dioxide and the Phosphoric acids contribute directly to the acid rain.
- Pollutants like lead, aluminum, methyl mercury, cadmium, etc. are not directly released by acid rain but triggered indirectly. Increased concentration of lead in drinking is especially harmful for children and infants.
- **Aluminum** can lead to dialysis dementia, a disorder of the central nervous system, and harm people with impaired kidney functions. **Methyl mercury** accumulates in the edible tissues of the fish and harms humans whereas **cadmium** can cause the corrosion of the galvanized pipes and enter the drinking water.

Q.34) Ans: C

Exp:



- **Thermal pollution** is caused by the release of the hot and cold water into the water bodies from the factories or the power plants generating electricity. Both of them have adverse ecological impacts on the water bodies they are released to.
- Hot water can cause alteration in the life supporting enzyme activity in aquatic life. This decreased enzyme activity can cause reduction in the breakdown of lipids and hence cause malnutrition in the animals.
- Warmer waters cause an increase in the growth of plants leading to greater population density and reduced light intensity. This further leads to decreased photosynthetic activity in the plants. Higher water temperature causes greater metabolism in the animals as they consume more food leading to food shortages in aquatic life.
- Higher temperature causes the increase in the population of the unwanted blue green algae while at the same time, it reduces the population of the green algae. Increased temperatures also lead to migration of organisms from the areas of less adaptability to that of greater adaptability. Hence, it causes the migration of the organisms from one place to another.

Q.35) Ans: C

Exp:

- **Statement 1 is correct:** **Plastic pollution** is highly pronounced in the marine environment and accounts for 70% of the marine pollution. The means of retrieval, sorting, and removal of plastics is difficult in the ocean environment as compared to the land. The rate of ultraviolet degradation of the plastic is more pronounced in the land than in the oceans. With the same amount of solar light exposed to the sea and land, the degradation of plastics in sea is much slower than land.
- **Statement 2 is correct:** Traditionally, plastics are regarded as bio-inert and have no toxic effect on aquatic plants directly. The addition of plastic to the marine environment can cause physical obstruction of the organisms and may also impact them physiologically. For example: the ingestion of Plastics by the fishes can cause fullness in them. The toxic and non-toxic organic matters present in the water at lower concentrations are enhanced with the addition of plastics to the marine environment.

Q.36) Ans: B

Exp:

- **Statement 1 is incorrect:** **Sanitary landfills** are the improved version of traditional landfills which are lined with the impermeable materials like plastics, clay, etc. in order to reduce the leaching of water and hence of the toxic and contaminants into the soil. The construction of these is very expensive although they prevent the problems of leaching as witnessed in the traditional landfills.
- **Statement 2 is incorrect:** In **vermiculture**, earthworms are added to the compost and they bring about the breakdown of the wastes. The excreta of the earthworms also enhances the nutrient concentration of the wastes. Hence, this method is also known as earthworm farming.
- **Statement 3 is correct:** **Pyrolysis** is a method of burning the materials in the absence or in the controlled environment of oxygen. It is used as an alternative to incineration. The products obtained as a result of pyrolysis in the form of solids and liquids are used as fuels. The products of pyrolysis include tar, charcoal, acetone, acetic acid, flue gas, etc.
- **Statement 4 is incorrect:** **Incineration** is not an environment friendly method for the burning of the wastes. In this process, the wastes are burnt at very high temperatures releasing huge amounts of toxic and polluting materials to the environment.

Q.37) Ans: C

Exp:

- **Composting** is the method of degradation of wastes in the presence of microorganisms like bacteria and fungi. It is done to decompose the degradable organic wastes. It is **done in the presence of oxygen**. The end product of composting is rich in carbon and nitrogen and serves as an excellent medium for the growth of plants. It is clean, safe, and cheaper. It also enhances the moisture retaining capability of the soil and makes it possible for the soil to hold greater nutrients.



Q.38) Ans: C

Exp:

- **Statement 1 is correct:** Alpha, Beta, Gamma particle radiations are released due to the process of radioactivity, where there occurs disintegration of an atomic nuclei into other elements. Above three are the particle radiations emitted as a result of radioactivity. **Alpha particles** can be blocked by the human skin, **Beta particles** however can penetrate into the skin although they can be blocked by materials like glass, metals, etc. **Gamma radiation** penetrates deep into the skin and brings about damage to the cells. They can be blocked only with a very thick and strong concrete material and hence cause huge damage to the humans.
- **Statement 2 is correct:** Radiations can be classified as **ionizing** and **non-ionizing** depending upon their ability to get absorbed and penetrate into the objects. **Ionizing radiation** has huge penetrating power and hence can cause the breakdown of macromolecules. Ultra violet radiation which is released from the sun is short wave radiation which is non-ionizing in nature. They have low penetrating power but cause damage to the cells that absorb them. They cause severe damage to the eyes, skin, etc. when being absorbed.

Q.39) Ans: A

Exp:

- **Statement 1 is incorrect:** **Central Ground Water Authority** under the **Ministry of Jal Shakti** was created under the provisions of the Environment (Protection) Act, 1986. It is a National Apex Agency tasked with providing scientific inputs for the management, exploration, restoration, augmentation, etc. of the ground water resources of the country. **The Central Water Commission** is a premier technical organization of the country tasked with the responsibilities for assessing, coordinating, and consulting the concerned state governments for the conservation, protection, and utilization of the water resources of the country. It functions as an attached office under the Ministry of Jal Shakti.
- **Statement 2 is correct:** **International Convention for the Prevention of Pollution from Ships** is the main international convention that seeks to prevent the pollution from ships to the marine

environment on account of the operational and accidental causes. It was adopted at the **International Maritime Organization** in 1973 and is also known as the **MARPOL convention**. India is a signatory to the above convention.

Source: <http://cgwb.gov.in/aboutcgwb.html> ;
<http://www.cwc.gov.in/> ;
[https://www.imo.org/en/About/Conventions/Pages/International-Convention-for-the-Prevention-of-Pollution-from-Ships-\(MARPOL\).aspx](https://www.imo.org/en/About/Conventions/Pages/International-Convention-for-the-Prevention-of-Pollution-from-Ships-(MARPOL).aspx)

Q.40) Ans: A

Exp:

- **Statement 1 is incorrect:** The **E-waste Management Rules, 2016** under the Ministry of Environment, Forests, and Climate Change are applicable to all the e-wastes and the electrical and electronic components that are listed in its Schedule I. Hence, they also mention the Mercury based CFLs. These rules on the other hand do not apply to the Lead acid batteries, Micro enterprises, and the radioactive wastes.
- **Statement 2 is incorrect:** **Extended Producer Responsibility** refers to the responsibility of any producer of electrical or electronic equipment to ensure environmentally sound management of such waste. 'Extended Producer Responsibility - Authorisation' means a permission given by the Central Pollution Control Board to a producer, with clear implementation plans and targets outlined in such authorization for the management of the e-wastes. It may involve processes like take-back or the setting up of the collection centres for the management of such wastes.
- **Statement 3 is correct:** The rules are applicable to every manufacturer, consumer, bulk consumer, collection centres, retailers, dealers, dismantlers, etc. Hence, they include both the producers and the consumers.

Source: <https://greene.gov.in/wp-content/uploads/2018/01/EWM-Rules-2016-english-23.03.2016.pdf>

Q.41) Ans: D

Exp:



- The **Waste to Wealth** Mission of the **Prime Minister's Science, Technology, and Innovation Advisory Council** deals with methods to identify, develop, and deploy technologies that aim to treat wastes to generate energy, recycle them, and to extract value from them. The mission hence seeks to achieve the goals of a clean and green environment.
- The mission also aims to assist the Smart Cities and Swachh Bharat programmes and hence create economic models that are financially viable for the management of the wastes.
- The method of **Plasma Arc gasification** can be used to generate high temperature plasma arc and convert the waste into syngas and the residue ash.
- The syngas can be used further for the generation of electricity while the residue ash can be mixed with the cement and used for construction purposes. The dry and combustible wastes including the polymers can be subjected to **incineration, gasification, and pyrolysis**, etc.
- The heating of the wastes at high temperature in case of pyrolysis can lead to the generation of syngas and pyrolysis oil that can be further used for the heating purposes.
- Biomass wastes like dry leaves and grasses can be mixed with the slurry in the biogas digester plants and be used for the production of cooking fuel and the production of syngas.

Source: <https://www.psa.gov.in/mission/waste-wealth/38> ;
<https://pib.gov.in/PressReleasePage.aspx?PRID=1667099>

Q.42) Ans: C

Exp:

- **Statement 1 is correct:** **Organochlorines** are the chlorinated hydrocarbon compounds that belong to the class of Persistent Organic Pollutants (POPs) and widely used as pesticides (nearly 40% in Asia). They cause huge adverse impacts on the Central Nervous Systems. DDT, Methoxychlor, chlordane, etc. are the group of compounds that belong to this.
- **Statement 2 is correct:** **Stockholm convention** is a global treaty that seeks to prevent human health from the harmful effects of the Persistent Organic Pollutants (POPs) that persist in the

environment, bio-accumulate in the living organisms and hence cause huge environmental damage. India had ratified the Stockholm Convention in 2006. The Ministry of Environment, Forest and Climate Change (MoEFCC) notified the **'Regulation of Persistent Organic Pollutants Rules, in 2018** under the provisions of the Environment (Protection) Act, 1986. The regulation prohibited the manufacture, trade, use, import and export of seven chemicals.

Source:

<https://dhss.delaware.gov/dhss/dph/files/organochlorpestfaq.pdf> ;
<https://pib.gov.in/PressReleasePage.aspx?PRID=1662335>

Q.43) Ans: A

Exp:

- **Statement 1 is incorrect:** **Hydrogen fuel cells** are responsible for the generation of electricity using chemical reactions from one electrode to another using an electrically charged electrolyte. It also involves the use of catalysts that speeds up the process of energy generation. Continuous charging is not required in the case of hydrogen fuel cells that provide electricity as long as it is provided with the hydrogen and oxygen. Unlike batteries, periodic charging is not required in this case.
- **Statement 2 is incorrect:** The limitations of hydrogen fuel cells include the higher cost of the catalyst used, problems associated with the maintenance and the storage of the hydrogen. Usually, platinum is used as a catalyst. It is expensive and makes the hydrogen fuel cell expensive.
- **Statement 3 is correct:** A chemical reaction taking place in the apparatus is responsible for the generation of the electric energy. It is composed of electrically charged electrolytes and the electrodes in the form of anode and cathode where the movement of charge from one electrode to another leads to the generation of the electrical energy.

Source:

<https://indianexpress.com/article/explained/hydroge>



[n-cars-electric-vehicle-supreme-court-how-cars-can-run-on-hydrogen-6168796/](#)

Q.44) Ans: A

Exp:

- **Statement 1 is correct:** Sustainable Alternative towards Alternative Transportation or SATAT was launched to provide an ecosystem for the production of Compressed Biogas from various biomass sources in the country. It envisages the setting up of 5000 Compressed Biogas Plants by 2023-24 and was launched by the Ministry of Petroleum and Natural Gas. The generated biogas would be made available to the automobiles in the country in the form of fuel.
- **Statement 2 is incorrect:** One sun, one world, one Grid is an initiative of the Ministry of New and Renewable Energy and was mooted in 2018 in the General Assembly of the International Solar Alliance or ISA. The initiative aims to build a global ecosystem of interconnected renewable energy resources by 140 countries with India at the center. It covers two zones i.e. South Asia and South East Asian countries in the East and the Middle East and the African countries in the west.

Source:

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=1723513> ; https://www.business-standard.com/article/current-affairs/one-sun-one-world-one-grid-all-you-need-to-know-about-solar-strategy-120081500417_1.html

Q.45) Ans: A

Exp:

- **Option A is correct:** Electric Vehicles are regarded as a better alternative to the conventional vehicles as they are environment friendly. However, they require vehicle charging points at several locations to enable them to function without interruption. They are powered through rechargeable batteries present inside the car.

- They offer several advantages over conventional vehicles. However, there are certain limitations as well. They **require longer charging time**. The government aims to set up at least one electric vehicle charging kiosk at around 69000 petrol pumps in order to ensure greater electric mobility for the people and the faster adoption of electric cars.
- It is free from environmentally damaging pollutants and operates with minimum noise. The limitations however include **greater pricing of the cars** along with their ability to travel to shorter distances.

Source: <https://www.conserve-energy-future.com/advantages-and-disadvantages-of-electric-cars.php>

Q.46) Ans: D

Exp:

Directly or indirectly light affects the plants life in various ways. Plants that grow best in full sunlight are known as **heliophytes**, whereas plants that grow best in lower light intensities are known as **sciophytes**.

Point 1 is correct: Most of the plants except few, like seedlings of conifers, young fronds of ferns, some mosses and algae, need light for chlorophyll formation. Thus the existence of other life forms, which depends upon plants for food is due to light.

Point 2 is correct: Overall vegetative development of the plant parts are dependent upon light source. There are various adaptations among plants with respect to growth in different light intensities, like different morphological and physiological features such as :thicker stems, frequent branching, longer roots, root to shoot ratio etc

Point 3 is correct: Indirectly light affects transpiration rates through increase in temperature. Transpiration rates correspondingly affect water absorption also. Thus high light intensities are always associated with dry habitats and high transpiration rates.

Point 4 is correct: Opening and closing of stomata is influenced by light. This is thus related to transpiration and absorption



Point 5 is correct: Light conditions at poles are different from other parts of earth. Thus the total amount of radiation received by the earth's surface differs with latitude (distance from the equator). This may be one of the reasons for differences in the vegetation at poles and at other parts of the earth.

Q.46) Ans: D

Exp:

Statement 1 is correct: It is known as trophic pyramid as it explains the flow of energy from one trophic level to the next. Pyramid of energy will never appear inverted and is always upright because the energy stored in one source is always lost upon transfer. They are expressed in units of energy per area per time.

Statement 2 is correct: Only about 10% of the energy available to one trophic level gets transferred to the next trophic level. Each level should be roughly one tenth of the size of the preceding level (as energy transformations are around 10%). The bottom level will always represent the producers, with subsequent levels representing consumers (primary, secondary etc.). Ecological efficiency can be defined as the amount of energy at each trophic level in proportion to the next trophic level and it is the only ecological pyramid that explains the metabolic losses.

Statement 3 is correct: Since energy cannot be recycled and there is successive reduction in energy flow at successive trophic levels, the shorter the food chain, more will be the available food energy (for e.g. producer-primary consumer). The reason is with an increase in the length of the food chain, there are more trophic levels and corresponding more loss of energy; **10% rule.**

Q.47) Ans: A

Exp:

Pair 1 is correctly matched: Bergmann's Rule: Birds and mammals obtain greater body size in cold regions than in warm areas. But poikilotherms are smaller in cold regions. The tail, snout, ears, and legs of mammals are relatively shorter in colder parts than in warmer areas.

Pair 2 is correctly matched: Allen's Rule: Animals adapted to cold climates have shorter limbs and bodily appendages than animals adapted in warm climates. More specifically, it states that the body surface area to volume ratio for homeo-thermic animals varies with the average temperature of the habitat to which they are adapted.

Pair 3 is correctly matched: Rensch's rule: The races of birds with relatively narrow and more acuminate wings tend to occur in colder regions, while those in warmer regions tend to be broader.

Pair 4 is correctly matched: Jordan's Rule: Temperature affects the morphology of some fishes and is found to have relationships with the number of vertebrates.

Pair 5 is correctly matched: Gloger's Rule: It is an ecogeographical rule which states that within a species of endotherms, more heavily pigmented forms tend to be found in more humid environments e.g.- near the equator. It is found birds in more humid habitats tended to be darker than their relatives from regions with higher aridity.

Q.48) Ans: A

Exp:

Cells are the smallest unit of life, and hence are often referred to as the "building blocks of life". Cells consist of cytoplasm enclosed within a membrane, which contains many biomolecules such as proteins and nucleic acids. **Protoplasm** can be defined as the organized colloidal substances (such as proteins and water) that constitute the living nucleus, cytoplasm, plastids, and mitochondria of the cell.

Tissues are groups of cells that have a similar structure and act together to perform a specific function.

An **organism** refers to a living thing that has an organized structure, can react to stimuli, reproduce, grow, adapt, and maintain homeostasis. An organism would, therefore, be any animal, plant, fungus, protist, bacterium, or archaeon on earth.



Population is a number of all the organisms of the same group or species who live in a particular geographical area and are capable of interbreeding.

A **community** is a group of living things with commonality of environment, interactions, biotic and abiotic factors.

Ecosystem is a set of organisms and abiotic components connected by the exchange of matter and energy.

A **biome** is a collection of plants and animals that have common characteristics for the environment they exist in.

Q.49) Ans: B

Exp:

Statement 1 is incorrect: The r- strategists support variations of biotic and abiotic conditions for each parameter, taking into account their low demands. Their ecological niche, which represents the hypervolume common to all these parameters, will therefore be very large.

Statement 2 is correct: The r-strategists species populations are governed by their biotic potential (maximum reproductive capacity). They are very comfortable in a new ecosystem because it meets all their needs, since there are no competitors.

Whereas, the **K-strategists** are located only on a very small part of the food spectrum and therefore rarely compete with each other. There are thus numerous species within a single ecosystem but the number of individuals is limited by the biological potential of their ecological niche.

The hypervolume that characterizes their specific ecological niche will be very small and they have limited function but they carry it out with maximum efficiency. The K-strategists have tiny ecological niches that are very narrow and close to one another, but each niche is occupied. It is because of this that a mature ecosystem is well balanced.

Q.50) Ans: a

Exp:

Option a is correct; According to Gause, it is acknowledged that two species that possess the same ecological niche cannot live together in the same ecosystem. This is the principle of reciprocal exclusion, which is linked to competition. If this principle seems true in the absolute, it will nevertheless probably be inapplicable in nature because there are a number of dimensions of niches in an ecosystem.

Option b relates to Fundamental niche; Each animal or plant species having its own vital preferences can develop very well within its adaptive potentialities, taking into account competitions. In consequence, its ecological niche corresponds to the hypervolume present in figure. This hypervolume is called fundamental niche.

Option c relates to Realized niche; Most frequently, abiotic conditions, and above all interspecific competition, ensure that the species does not express its entire genetic potential. It develops only a part of its potential. The hypervolume obtained in a given ecosystem is much smaller than what is possible.

Q.51) Ans: D

Exp:

Daily responses of animals to light conditions are known as circular rhythms, whereas the annual rhythms known as circannual rhythms.

Statement 1 is incorrect: Circadian rhythms are internally driven or endogenous, are affected little by temperature changes, are sensitive to great variety of chemical inhibitors; and are innate, not learned from or imprinted upon the organism by the environment. The innate character of circadian rhythm is demonstrated by several animals.

Statement 2 is incorrect: The circadian rhythms have been observed in zoo planktons, polychaete annelids, many insects, most birds and some mammals. Planktons of sea and lakes also show diurnal changes in their vertical distribution.

Q.52) Ans: D



Exp:

Ecosystem is a set of organisms and abiotic components connected by the exchange of matter and energy, it is self sustained, functional unit and contains following structural factors:

Point 1 is correct: Species composition, It differs from one ecosystem to another depending upon geography, topography and climate. Maximum species composition occurs in tropical rain forests and coral reefs. Minimum occurs in deserts and arctic regions.

Point 2 is correct: Stratification, It is the formation of vertical layers where vegetation is dense. It is absent or rare in deserts.

Point 3 is correct: Trophic structure, Each ecosystem has specific food chains and food webs.

Point 4 is correct: Standing State, It is the amount of inorganic nutrients present in soil/water of the ecosystem.

Point 5 is correct: Standing crop, It is the amount of living biomass present in an ecosystem.

Q.53) Ans: B

Exp:

Diapause can be defined as the physiological state of dormancy where there is no development and most life processes are shut down. It is initiated during unfavorable conditions and can most commonly be observed in insects where the eggs and larvae undergo a resting period during unfavorable conditions.

Statement A relates to hibernation: Hibernation is the state of minimal activity and metabolic depression. Hibernation is a seasonal heterothermy characterized by low body temperature, slow breathing and heart rate, and low metabolic rate. It commonly occurs during winter months.

Statement C relates to aestivation: Aestivation is a state of animal dormancy, similar to hibernation, although taking place in the summer rather than the winter. Aestivation is characterized by inactivity and a lower metabolic rate that is entered in response to high temperatures and arid conditions.

Statement d relates to resting phase in cell cycle:

More appropriately called interphase. The interval in the cell cycle between two cell divisions when the individual chromosomes cannot be distinguished, interphase was once thought to be the resting phase but it is far from a time of rest for the cell.

Q.54) Ans: a

Exp:

MacArthur discovered that when two species with similar requirements coexist, each typically occupies a smaller niche than either would by itself. It is done to help avoid competition in an ecological niche. In any environment, organisms compete for limited resources, so organisms and different species have to find ways to coexist with one another. This results in specific physiological, morphological or behavioral adaptations, which is known as **character displacement**. Hence **Statement a is correct**.

Statement b is incorrect: Related to the concept of "rescue effect." It explains an increase in fitness of individuals on islands closer to the mainland through increased immigration rates. Local extinction rates of species on these islands are lower than on islands that are more isolated and farther away from the mainland.

Statement c is incorrect: Related to the concept of "Müllerian mimicry." It is a form of biological resemblance in which two or more unrelated noxious, or dangerous, organisms exhibit closely similar warning systems, such as the same pattern of bright colours.

Statement d is incorrect: Related to the concept of "allele effect." It is characterized by a correlation between population size or density and the mean individual fitness of a population or species.

Q.55) Ans: D

Exp:

- Statement 1 is correct:** Pyrolysis/gasification, Anaerobic Digestion, Combustion are the methods of converting biomass into other forms of energy. Biomass is a form of renewable energy resource and constitutes



organic wastes released by humans and their activities.

- **Statement 2 is correct:** Although burning of biomass releases some amount of pollutants like carbon dioxide, etc. However, it does not add any extra pollution to the environment. The carbon dioxide released by it on burning is the same as the amount absorbed by it during its formation.
- **Statement 3 is correct:** India's renewable energy target involves **175 GW** of energy from renewable sources by **2022**. The share of different sources are: **solar: 100 GW; Wind: 60 GW; Biomass: 10 GW; and small hydro power plants: 5 GW.**
- **Statement 4 is incorrect:** Burning of biomass does release carbon dioxide like the fossil fuels. It releases the approximately same amount of carbon dioxide as fossil fuels.

- **Statement 1 is incorrect:** Dissolved Oxygen in an aquatic ecosystem directly depends on the amount of the organic and inorganic wastes present in the water body. Oxygen is required by the aquatic species for the decomposition and consumption needs. The presence of greater amounts of wastes decreases the dissolved oxygen in the water body as the rate of its decomposition is increased. This leads to increased demand of oxygen by the aquatic species which is referred to as the Biological Oxygen Demand (BOD). BOD hence indicates the oxygen needed for the decomposition of the organic matter in the water body. More the BOD in a water, greater is the deficiency of Dissolved Oxygen (DO) and poor is the quality of the water. Chemical Oxygen Demand (COD) on the other hand measures the amount of oxygen required for the decomposition of both bio-degradable and non-biodegradable matter in the water body. Hence, COD is the better measure for the estimation of health of an aquatic system.

- **Statement 2 is incorrect:** COD measures the amount of oxygen required for the oxidation of both biodegradable and non-biodegradable matters in an aquatic system. Hence, it is a measure to indicate the health of an aquatic ecosystem.
- **Statement 3 is incorrect:** Greater the number of dissolved wastes in a water body, more is the requirement of oxygen for its decomposition which leaves an aquatic system with greater demand for oxygen as most of the oxygen dissolved is used up in the decomposition of the wastes. Hence, the amount of dissolved oxygen is inversely proportional to the number of dissolved wastes in a water body.

Q.56) Ans: D

Exp:

- **Statement 1 is incorrect:** The Central Government launched the National Clean Air Programme (NCAP) as a long-term, time-bound, national level strategy to tackle the air pollution problem across the country. It targets to achieve 20% to 30% reduction in Particulate Matter concentrations by 2024 keeping 2017 as the base year. Under NCAP, 122 non-attainment cities (those that have fallen short of the National Ambient Air Quality Standards (NAAQS) for over five years) have been identified across the country based on the Air Quality data from 2014-2018. Air quality of cities is monitored by State Pollution Control Boards which publishes their results from time to time.

Statement 2 is incorrect: It is prepared by the **Ministry of Environment, Forests and Climate Change.**

Q.57) Ans: D

Exp:

Q.58) Ans: D

Exp:

- **Statement 1 is correct:** It is the responsibility of the waste generator to segregate into three categories: Biodegradable, non-biodegradable, and domestic hazardous wastes.
- **Statement 2 is correct:** It is the responsibility of the local authorities and the village



panchayats to make plans for the suitable disposal of the wastes as per the policy formulated by the state governments.

- **Statement 3 is correct:** Financial assistance is to be provided by the manufacturers of the disposal products to the local authorities for setting up of waste management facilities.
- **Statement 4 is incorrect:** The government has established a Central Monitoring Committee under the MoEF&CC to monitor the overall implementation of the rules. The rules are applicable to all local bodies and the census towns. It prohibits the construction of landfills in the hilly areas.

dictated not by total resources available, but by the scarcest resource (limiting factor).

- **Statement 2 is correct:** The base of the food web in the terrestrial ecosystem is occupied by Plants/Autotrophs.. They are the primary producers because they make their own food by converting sunlight through photosynthesis.
- **Statement 3 is correct:** The degradation of biodiversity, biodiversity loss, deterioration of soil quality, Potential changes in the distribution of the flora and fauna, change in ecosystem structure and function, and alterations in soil characteristics etc are caused by changes in land use.

Q.59) Ans: A

Exp:

- **Statement 1 is correct:** Bioremediation is the process of breakdown of environmental pollutants into less toxic forms by the microorganisms like the bacteria or the fungi, etc., or their enzymes.
- **Statement 2 is correct:** The National Institute of Ocean Technology (NIOT) has developed an eco-friendly crude oil bioremediation mechanism technology using consortia of marine microbes wheat bran (WB) immobilized on agro-residue bacterial cells.
- **Statement 3 is incorrect:** These hydrocarbon degrading bacteria do not depend on hydrocarbons for survival but have a metabolic mechanism of their own where they use petroleum products as carbon and energy source and thus, help cleaning up oil spills.

Q.61) Ans: A

Exp:

- **Statement 1 is correct:** Grazing food chain is a type of food chain in which energy at the lowest trophic level is acquired through the process of photosynthesis. Grazing food webs have a primary producer at their base, which is either a plant for terrestrial ecosystems or a phytoplankton for aquatic ecosystems. The producers pass their energy to the various trophic levels of consumers.
- **Statement 2 is incorrect:** Terrestrial ecosystems are affected by temperature, sunlight, the average amount of precipitation received, the soil type, etc. Whereas, the factors that affect the aquatic ecosystems are salinity, acidity, oxygen saturation, sunlight, depth, and temperature, water flow etc. Salinity is not a factor affecting the Terrestrial Ecosystem.
- **Statement 3 is incorrect:** In the aquatic ecosystems, most of the photosynthesis that occurs is carried out by the Phytoplanktons. Even though phytoplankton are small, they grow very rapidly and they support large populations of herbivores. This is the reason the aquatic ecosystems can support more trophic levels than the land ecosystems.

Q.60) Ans: D

Exp:

- **Statement 1 is correct:** A limiting factor is anything that constrains a population's size and slows or stops it from growing. Some of the limiting factors in a terrestrial ecosystem are space, temperature, altitude, and amount of sunlight available in an environment. According to Liebig's law, the limiting factor is the one that is the least available at a particular time (e.g. prey or nutrients). It states that growth is



Q.62) Ans: C

Exp:

- **Statement 1 is correct:** The changes in precipitation modify the amount and quality of habitat for aquatic organisms, interfere with the reproduction of many aquatic species, disrupt present patterns of plant and animal distribution and thus, they affect their hydrological characteristics ultimately influencing the ecosystem's productivity and diversity.
- **Statement 2 is correct:** Solubility of oxygen is a function of water temperature, salinity, and atmospheric pressure. It decreases with rising temperature and salinity, and increases with rising atmospheric pressure. Dissolved oxygen decreases exponentially as salt levels increase. That is why, at the same pressure and temperature, saltwater holds about 20% less dissolved oxygen than freshwater.
- **Statement 3 is incorrect:** Aphotic Zones are the deepest layer of the oceans and they get no sunlight at all. There is primarily no light from the sun therefore photosynthesis can not take place in this zone of the aquatic ecosystems.

Q.63) Ans: B

Exp:

- **Statement 1 is incorrect:** Aquatic ecosystems include oceans, lakes, rivers, streams, estuaries, and wetlands. Estuaries are coastal areas where the saline waters of the ocean meet with fresh water from streams and rivers. Estuaries offer good fishing grounds for the fisherman as the water column is shallow. They also maintain water quality through natural filtration as microbes break down organic matter and sediments bind pollutants.
- **Statement 2 is correct:** Monocropping is an agricultural practice that depletes the soil's nutrients. It weakens the biome and increases its vulnerability to natural disasters. Agricultural monoculture upsets the natural balance of soils. It can also lead to the buildup of pests and diseases

affecting crops and plants.

- **Statement 3 is correct:** Fungi are important decomposers in the aquatic ecosystems. The other important decomposers in marine ecosystems are bacteria, marine worms, crustaceans, mollusks etc. In the colder ocean waters, only bacteria and fungi decompose because the other creatures cannot survive in the extreme conditions.

Q.64) Ans: B

Exp:

- **Pair 1 is correct:** Eutrophication is a phenomenon when the aquatic environment becomes enriched with nutrients. It causes several structural changes to the ecosystem such as: increased production of algae and aquatic plants, depletion of fish species, general deterioration of water quality etc. The enrichment of water by nutrients can be natural or anthropogenic in nature.
- **Pair 2 is incorrect:** The limnetic zone is the open and well-lit area of a freestanding body of freshwater, such as a lake or pond. The zone is dominated by plankton and therefore it is the most photosynthetically-active zone of a lake/pond.
- **Pair 3 is correct:** Trophic cascade is an ecological phenomenon caused by the addition or removal of top predators which leads to changes in the relative populations of predator and prey through a food chain. This often results in dramatic changes in ecosystem structure and nutrient cycling. Example- When Tigers are reduced from an ecosystem, the prey biomass increases. This, in turn, could cause an increase and/or decrease in biomasses of other species throughout the food web.

Q.65) Ans: A

Exp:



- **Statement 1 is correct:** Nocturnality is a behavioural adaptation of many species of the desert Ecosystem to avoid the heat of the day. This behavior prevents animals from losing water by avoiding heat because the deserts are very hot during the daylight hours. This adaptation also enhances osmoregulation.

- **Statement 2 is incorrect:** The organisms that eat the producers are the primary consumers. In the Terrestrial Ecosystem, the primary consumers are the Herbivores. Other examples of primary consumers are Caterpillars, insects, grasshoppers, termites and hummingbirds, rabbits, giraffes, horses etc.

- **Statement 3 is correct:** Terrestrial ecosystems are one of the natural carbon sinks of the earth and contribute considerably to the global Carbon balance. Terrestrial carbon sequestration is the process through which CO₂ from the atmosphere is absorbed by trees and plants through photosynthesis and stored as carbon in soils and biomass.

Q.66) Ans: B

Exp:

- **Statement 1 is correct:** An ecotone is an area that acts as a boundary/ transition between two ecosystems. Examples of ecotones include mangrove forests (between terrestrial and marine ecosystems), grasslands (between desert and forest), and estuaries (between saltwater and freshwater). In a similar way, Wetlands are ecotones between Terrestrial and Aquatic Ecosystems.

- **Statement 2 is incorrect:** The boreal forests/Taiga forests are characterized by its diverse coniferous tree species. Conifers like Pine, Fir, Larch, Spruce, Hemlock are well adapted for harsh conditions to survive the long, cold winters and short summers of the taiga. Their needles contain very little sap, which helps prevent freezing.

- **Statement 3 is incorrect:** The tropical rainforest is a hot, moist biome where it rains all year long. It is known for its dense canopies of vegetation. The canopy blocks winds, rainfall, and sunlight, creating a humid, still, and dark environment below.

Q.67) Ans: C

Exp:

- **Pair 1 is correct:** The biodiversity of the tundras is low. Very few plants grow due to the lack of nutrients and because the ground is frozen beneath the top layer of soil. Low precipitation in the tundra region does not provide enough water to support trees. Instead, the tundra has patchy, low-to-ground vegetation consisting of small shrubs, grasses, mosses, sedges, and lichens, all of which are better adapted to withstand tundra conditions.

- **Pair 2 is incorrect:** A Lotic Ecosystem (also called the riverine ecosystem) has flowing waters. Examples include: creeks, streams, runs, rivers, springs, brooks and channels. A Lentic Ecosystem (also called the lacustrine ecosystem) has still waters. Examples include: ponds, basin marshes, ditches, reservoirs, seeps, lakes, and vernal / ephemeral pools.

- **Pair 3 is incorrect:** Algae are not plants, animals or fungi. They are single-celled, plant-like organisms. They are autotrophs that fulfill the role of producer in ecosystems because they make their own food via photosynthesis, like plants. Producers like algae form the basis of energy in a food web. They are not decomposers.

Q.68) Ans: B

Exp:

- **Statement 1 is incorrect:** Net primary productivity is gross primary productivity minus the rate of energy loss to metabolism and maintenance. The highest net primary productivity in terrestrial environments occurs in swamps and marshes and tropical rainforests; the lowest occurs in deserts.



- **Statement 2 is correct:** Thermal stratification is the phenomenon in which lakes develop two discrete layers of water of different temperatures with depth. This temperature difference causes the colder and heavier water to settle at the bottom while allowing the warmer and lighter water to float over the colder water. The warm layer on top is called epilimnion and the cold layer below is called hypolimnion.
- **Statement 3 is correct:** Estuarine environments are among the most productive on earth because of the mixing of freshwater and saline water zones. These environments create more organic matter each year than comparably-sized areas of forest, grassland or agricultural land.

Q.69) Ans: C

Exp:

- **Statement 1 is correct:** Greater biodiversity in ecosystems, species, and individuals leads to greater stability. For example, species with high genetic diversity are more likely to be able to withstand weather disturbances, disease, and climate change etc.
- **Statement 2 is correct:** An Ecotone describes an area that acts as a transition or boundary between two ecosystems. This area is influenced by the two bordering ecosystems, and therefore it has a higher density and variety of species within it. This increase in biodiversity is referred to as the "edge effect".
- **Statement 3 is incorrect:** The biotic factors affecting the Aquatic ecosystems include plants, animals, and microbes. Some of the important abiotic factors include the amount of sunlight in the ecosystem, the amount of oxygen and nutrients dissolved in the water, proximity to land, depth, and temperature.

Q.70) Ans: A

Exp:

Biological oxygen demand(BOD) is amount of dissolved oxygen needed by bacteria for decomposing the organic waste present in water it is expressed in Milligram of oxygen per litre of water.

Sewage and untreated water discharge results in a decreased amount of body as much of the available dissolved oxygen is consumed by aerobic bacteria in the degradation process robbing other aquatic animals of oxygen they need to live. **Hence statement 1 is correct.**

BOD is one of the most common measures of positive organic material in a water body indicating the amount of organic matter present in water therefore lobia is an indicator of good quality water while high beauty indicates polluted water. Therefore clean water has less pollution than polluted water. **Hence statement 2 is incorrect.**

Source- previous year UPSC paper of 2017

Q.71) Ans: B

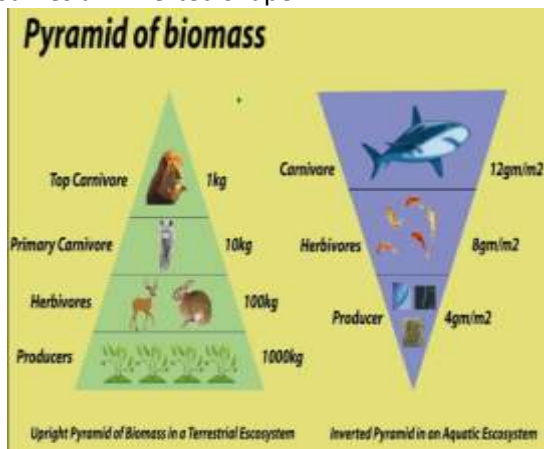
Exp:

- **Statement 1 is correct:** Blue carbon is the carbon stored and sequestered in coastal ecosystems such as mangrove forests, seagrass meadows or intertidal salt marshes. Whereas, the Green carbon is the carbon that is taken up by terrestrial ecosystems. It is the carbon stored in the plants and soil of natural ecosystems.
- **Statement 2 is correct:** The Suspended sediments from the nearby urban construction, agriculture and various other man-made activities, absorb heat energy which raises the water temperatures of the aquatic ecosystem.. Turbidity can also reduce light transmission through the water and decrease photosynthesis by aquatic plants, consequently affecting dissolved oxygen levels.
- **Statement 3 is incorrect:** Both natural and manmade wetlands have water purification functions. Natural wetlands purify our water by trapping sediments and removing pollutants from water. The Artificial wetlands are designed similar to natural wetlands to treat wastewater. They make use of the natural purification processes of vegetation, soils and microbes to remove contaminants from discharge.

Q.72) Ans: C

Exp:

- **Statement 1 is correct:** In the marine Ecosystems biomass pyramids are often inverted. This is because the producers are very small and have limited mass i.e, the biomass of fishes far exceeds the biomass of phytoplankton. The consumer biomass at any instant exceeds the producer biomass. The producers also reproduce and die quickly, so there is less biomass at any given time compared to consumers and hence the pyramid assumes an inverted shape.



- **Statement 2 is incorrect:** The Pyramid of energy represents the amount of energy at each trophic level of the food chain. It is the only type of ecological pyramid, which is always upright because the energy flow in a food chain is always unidirectional. Energy goes on decreasing as we move towards apex from base.



- **Statement 3 is incorrect:** Fungi do not have chlorophyll and cannot make their own food through photosynthesis. They are not the primary producers of an ecosystem. They are the decomposers that break down dead organisms

and waste materials into nutrients usable by the producers.

Q.73) Ans: D

Exp:

- **Statement 1 is correct:** When dissolved oxygen levels fall below 3 ppm many consumers, such as fish and zooplankton die. Hydrostatic pressure due to the water column also increases with depth in oceans. Animals in the deeper layers are adapted to the high pressure.
- **Statement 2 is correct:** Because water has a much higher heat capacity, or specific heat, than do sands, soils or other materials, for a given amount of solar irradiation (insolation), water temperature will increase less than land temperature. The land has a smaller "heat capacity" than water, which means it needs less heat to raise its temperature.
- **Statement 3 is correct:** Bioluminescence occurs through a chemical reaction that produces light energy within an organism's body. Through a process of natural selection, some organisms have developed the ability to enhance light production through physiological, molecular, anatomical, and behavioral adaptations. This bioluminescence is used to hunt prey, defend against predators, to confuse attackers, attract and find mates, etc.

Q.74) Ans: A

Exp:

- **Statement 1 is incorrect:** Some plants that are adapted to dry environments, use the CAM pathway to minimize photorespiration. CAM plants stomata only open at night, when humidity tends to be higher and temperatures are cooler, both factors that reduce water loss from leaves. They also absorb and store carbon dioxide in the night time. During the day, the stomata stay closed while the plant uses sunlight to convert the stored carbon dioxide into energy. Thus, minimizing water loss.
- **Statement 2 is correct:** Primary productivity rates within a desert ecosystem tend to be less than other ecosystems. The lowest net primary productivity occurs in deserts. These low levels are due to the very small amounts of precipitation/ rainfall and extreme temperatures.



- **Statement 3 is correct:** Xerophytes are those plants that grow in hot and dry regions such as deserts. Xerophyte plants have fewer stomata to reduce water loss via transpiration. The stomata may also be surrounded by tiny hairs on the leaf or stem. These hairs slow down the rate of transpiration as it shelters the stomata from harsh winds.

Q.75) Ans: C

Exp:

- **Statement 1 is correct:** Fires are a natural part of the grassland ecosystem and help maintain its health and vigor. It is believed that occasional fires help to rid the land of old grasses and allow for new grasses to grow, bringing new life to the area. Fires also warm up the soil and reduce the leaf litter that accumulates each year, allowing sunlight to penetrate.
- **Statement 2 is incorrect:** Buttress roots are roots that grow out from the tree trunk. They are most commonly found in the Tropical rainforest trees. These roots provide extra stability to the plants. They also share the weight of the tree and such roots from nearby trees might also intertwine creating an intricate mesh that helps support several trees and also helps in the nutrient uptake of the plants.
- **Statement 3 is correct:** The hot and humid conditions make tropical rainforests an ideal environment for bacteria and other microorganisms. Because these organisms remain active throughout the year, they quickly decompose matter on the forest floor. The high diversity of decomposers, such as bacteria and fungi also accelerates the decomposition process.

Q.76) Ans: D

Exp:

- **Statement 1 is incorrect:** Warm-blooded animals/Homeotherms are those animals that can regulate and maintain a constant internal body temperature. They can survive in any temperature range as they can adapt to it easily. The Warm Blooded Animals dominate The Terrestrial Environment. Examples- Mammals, Birds etc. However, there are several terrestrial living organisms that are cold blooded in nature. Example- Amphibians, reptiles etc.

- **Statement 2 is incorrect:** Hydrophytes are those plants which live in water. Hydrophytes generally have small and short roots. The main function of hydrophytic roots is for anchorage in the water. Roots are not so much required for absorption of water as leaves can absorb water through osmosis.
- **Statement 3 is incorrect:** Aquatic animals generally breathe air or extract oxygen dissolved in water through gills, or directly through the skin. However, Whales and dolphins are 2 aquatic animals that breathe through their lungs. They come up to the surface, inhale and once again submerge into the water.

Q.77) Ans: D

Exp:

- **Statement 1 is incorrect:** An Ecotone is a zone of junction or a transition area between two biomes.
- **An ecotype is a population (or subspecies or race)** that is adapted to local environmental conditions.
- **Statement 2 is incorrect:** Ecotones is the transition area between ecosystems. These areas experience the Edge Effect. They contain species from both bordering ecosystems, so they often contain greater biodiversity and density of organisms as compared to surrounding ecosystem. They also have significantly greater productivity as resources from both ecosystems can be accessed easily in the one place.
- **Statement 3 is correct:** Areas experiencing edge effects might have higher predation levels than interior habitats due to the following reasons- The predator activity is higher in areas with higher prey density and Predators are also more abundant on edges than in forest interior.

Q.78) Ans: A

Exp:

- **Statement 1 is correct:** Transpiration is the process that occurs when water is absorbed by roots from the soil and transported as a liquid to the leaves via xylem. In the leaves, stomata allows water to escape as a vapor. Transpiration happens in all plants with a vascular system, including aquatic plants.
- **Statement 2 is incorrect:** The intertidal zone is the area where the ocean meets the land between high and low tides. It is submerged with

water during high tide and exposed to the air during low tide. These zones experience edge effects as Intertidal zones are transitional coastal regions. These zones provide a home to specially adapted marine plants and animals.

- **Statement 3 is correct:** Podzols are found in high-latitude coniferous forest belts having very long cold winters, short cool summers and moderate precipitation throughout the year. These soils are acidic, infertile and of limited value for agriculture due to the sandy portion, resulting in a low level of moisture and nutrients. These are useful for dairying and mixed farming. Most Podzols are poor soils for agriculture.

Q.79) Ans: C

Exp:

- **Statement 1 is correct:** Mangroves face challenges in seed germination due to the presence of high concentration of salt in water. These plants solve this problem through vivipary. Vivipary is a type of reproductive adaptation shown by mangroves where seeds or embryos begin to develop before they detach from the parent. The embryo gets nourishment from the parent plant and grows out of seed in the form of a green seedling.
- **Statement 2 is incorrect:** The pelagic zone is the region of a body of water (lake, river, or ocean) that does not come in contact with the bottom or the shore.
- **Statement 3 is incorrect:** There is an inverse relationship between the dissolved oxygen and temperature. As the temperature of the water increases, dissolved oxygen levels decrease. The thermal pollution causes decrease in the dissolved oxygen. This can create suffocation and death of the aquatic flora and fauna.

Q.80) Ans: D

Exp:

- **Statement 1 is correct:** The tidal/mangrove forests are found in the deltas of all major rivers – Ganga, Brahmaputra Delta (the Sunderbans), the deltas of the Mahanadi, the Godavari, the Krishna, and the Kaveri. The highest concentration of tidal forests in India is found in the Ganges–Brahmaputra delta region.
- **Statement 2 is correct:** Kharai Camels or Swimming Camels are found only in Gujarat's

Bhuj area. This camel is adapted to the extreme climate of Rann of Kutch where shallow seas and high salinity is prevalent. The Kharai camels of Kutch are dependent on mangroves for their food. Thus, the destruction of the mangroves affects their food source, causing their numbers to decrease eventually.

- **Statement 3 is correct:** Mangrove trees are adapted for survival in oxygen-poor or anaerobic conditions through specialized root structures called pneumatophores/breathing roots. These roots extend upward from the underground roots above the soil surface to obtain Oxygen. The pneumatophores as an adaptive strategy to survive in a hypoxic environment

Q.81) Ans: B

Exp:

- **Food Waste Index Report 2021 published by the UN Environment Programme (UNEP) and partner organisation WRAP in March 2021.**
- **The Food Waste Index Report aims at supporting the goals of SDG 12.3.** It does so by presenting the most comprehensive food waste data collection, analysis and modelling to date, generating a new estimate of global food waste; and publishing a methodology for countries to measure food waste, at household, food service and retail level, to track national progress towards 2030 and to report on SDG 12.3.
- **An estimated 931 million tonnes of food, or 17 per cent of total food available to consumers in 2019, went into the waste bins of households, retailers, restaurants and other food services, according to the**
- Sustainable Development Goal (SDG) target 12.3, which aims at halving per capita global food waste at the retail and consumer levels and reducing food losses along production and supply chains, including post-harvest losses.
- **WRAP (Waste and Resource Action Programme)**
 - It is a British registered charity.
 - It works with businesses, individuals and communities to achieve a circular economy, by helping them reduce waste, develop sustainable products and use resources in an efficient way was established in 2000 with a



vision of a world in which resources are used sustainably.

- It's mission is to accelerate the move to a sustainable, resource-efficient economy

Q.82) Ans: B

Exp:

- The greater one-horned rhino (or "Indian rhino") is the largest of the rhino species. Once widespread across the entire northern part of the Indian subcontinent, rhino populations plummeted as they were hunted for sport or killed as agricultural pests.
- Nearly 85% of the global Indian rhinoceros population is concentrated in Assam, where Kaziranga National Park contains 70% of the rhino population.
- The other places where Rhino population are found within India are as follows:

- Dudhwa National Park and Katarniaghat Wildlife Sanctuary in the state of Uttar Pradesh
- Jaldapara National Park and Gorumara National Park in the state of West Bengal
- Pobitora Wildlife Sanctuary, Orang National Park and Manas National Park in the state of Assam.
- Outside India it is found in Parsa National Park, Chitwan National Park, Bardia National Park, Shuklaphanta Wildlife Reserve of Nepal.

- Protection Status of Greater One- Horned Rhinoceros also known as Indian rhinoceros (*Rhinoceros unicornis*)

- IUCN Red List: **Vulnerable**
- Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) : **Appendix 1**
- Wildlife Protection Act : **Scheduled 1**

Main threat to one horned rhino:

- Poaching for the horns as used in traditional Chinese Medicine.
- Habitat loss
- Population density
- Decreasing Genetic diversity

Conservation programs:

- Indian Rhino Vision (IRV) 2020 programme

- The program was launched in 2005 by Assam Forest Department in partnership with WWF-India, the International Rhino Foundation, and US Fish & Wildlife Service.
- It aims to attain a wild population of at least 3,000 greater one-horned rhinos spread over seven protected areas in Assam by the year 2020.

• New Delhi Declaration on Asian Rhinos 2019:

- Under the declaration, India will collaborate with Bhutan, Nepal, Indonesia, and Malaysia for conservation and protection of Asian rhino species

• Other Rhino Species:

- Southern White Rhino: Near Threatened in IUCN Red List.
- Northern White Rhino: Functionally extinct.
- Black Rhino: Critically Endangered in the IUCN Red List.

Q.83) Ans: B

Exp:

- The SVAMITVA scheme is not being implemented in Delhi as there are no rural local bodies in Delhi and villages of Delhi are under the administrative jurisdiction of municipalities.

States/Union Territories need to sign Memorandum of Understanding (MoU) with Survey of India for implementation of the SVAMITVA scheme. So far, 26 States/UTs have signed MoU.

- Department of Land Resources is implementing Digital India Land Records Modernization Programme (DILRMP), the erstwhile National Land Records Modernization Programme to digitize and modernize land records. The scheme was launched in 2008-09 as a Centrally Sponsored Scheme and revamped as a Central Sector Scheme with effect from 1st April, 2016 with 100% funding by the Government of India. The Scheme would be implemented across the country in phased manner during 2021-2025 by the union Ministry of Panchayati Raj.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1739567>



Q.84) Ans: B

Exp:

- **Statement 1 is incorrect:** Under Section 5 (Power to regulate place and conditions of detention) of THE NATIONAL SECURITY ACT, 1980-
—Every person in respect of whom a detention order has been made shall be liable—
 - (a) to be detained in such place and under such conditions, including conditions as to maintenance, discipline and punishment for breaches of discipline, as the appropriate Government may, by general or special order, specify; and
 - (b) to be removed from one place of detention to another place of detention, whether within the same State or in another State, by order of the appropriate Government: Provided that no order shall be made by a State Government under clause (b) for the removal of a person from one State to another State except with the consent of the Government of that other State.
- **Statement 2 is correct:** Under Section 3(Power to make orders detaining certain persons) of THE NATIONAL SECURITY ACT, 1980- the District Magistrates and Commissioner of Police can exercise the provisions of NSA. Section 3 (3) reads as follows-If, having regard to the circumstances prevailing or likely to prevail in any area within the local limits of the jurisdiction of a District Magistrate or a Commissioner of Police, the State Government is satisfied that it is necessary so to do, it may, by order in writing, direct, that during such period as may be specified in the order, such District Magistrate or Commissioner of Police may also, if satisfied as provided in sub-section (2), exercise the powers conferred by the said sub-section.

Q.85) Ans: D

Exp:

- The Government has unveiled its vision for the next decade, listing thereby ten most important dimensions in 2030. These are in sync in making India a modern, technology

driven, high growth, equitable and transparent society.”

- The dimensions of Vision-2030 are as follow:
 - To create physical and social infrastructure for a ten trillion dollar economy and to provide ease of living.
 - Digital India led by the youths with innumerable start-ups and millions of jobs.
 - To make India pollution free by focusing on Electrical Vehicles and renewables.
 - Rural industrialisation using modern technologies to generate massive employment.
 - Clean rivers, with safe drinking water to all Indians and efficient use of water in irrigation using micro-irrigation techniques.
 - Besides scaling up Sagarmala, India's coastline and ocean waters will power development
 - Through our space programme – Gaganyaan, India becoming the launch-pad of satellites for the World
 - Self sufficiency in food production and producing food in the most organic way.
 - A healthy India by 2030 and a distress free health care and wellness system for all. Ayushman Bharat and women participation would be an important component in it.
 - Employees working with elected Government, transforming India into Minimum Government Maximum Governance nation.

Q.86) Ans. A

Exp.

- NISHTHA: National Initiative for School Heads and Teachers Holistic Advancement (NISHTHA) is a national mission to improve learning outcomes at the elementary level. It was launched under the Centrally Sponsored Scheme of Samagra Shiksha by the Union Human Resource Development (HRD) Minister in 2019-2020 to build capacities of 42 Lakh government teachers across the country.



- **Statement 1 is correct** : In the first batch, a total of 120 School Principals and teachers from 26 EMRSs covering three states of Madhya Pradesh, Chhattisgarh, and Himachal Pradesh successfully completed the program. The integrated teacher training covered different aspects of education such as Learning Outcomes, School Based Assessments, POCSO Act, Gender Sensitization, Health and Well-Being in Schools, Teaching of Mathematics, Language, Science, and Social Science.
- **Statement 2 is incorrect**: National Education Society for Tribal Students (NESTS) (an autonomous body under the Ministry of Tribal Affairs (MoTA) has recently collaborated with National Council of Educational Research and Training (NCERT), which is a national apex education body for several programmatic initiatives and interventions, for conducting NISHTHA programs across 350 functional (Eklavya Model Residential Schools) EMRS schools across the country.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1728867>

Q.87) Ans: C

Exp.

- **Statement 1 is incorrect**: India has moved up a position to the third spot in the recently released Renewable Energy Country Attractiveness Index (RECAI). The improvement in performance is due to the exceptional performance on the solar PV front. Installed solar PV capacity in India has skyrocketed to 39 GW marginally overtaking the wind capacity for the first time.
- **Statement 2 is incorrect** : The index is released by Ernst & Young Global Limited, doing business as EY, is a multinational professional services network with headquarters in London, England. EY is one of the largest professional services networks in the world.
- Renewable Energy Country Attractiveness Index finds that environment, sustainability and governance (ESG) goals are increasingly being prioritized on the investor agenda while institutional investors' interest in renewables continues to grow.

- **Statement 3 is correct**: US retains the top position with China and India bringing up the top 3 in the rankings. In 2020, global renewable energy capacity investments grew 2% to US \$ 303.5b, the second-highest annual figure recorded to date despite the impact of the global COVID-19 pandemic.

Q.88) Ans. D

Exp.

- The National Board for Wildlife (NBWL) is a statutory body constituted by the Central Government under Section 5A of the Wildlife (Protection) Act, 1972. It is advisory in nature and advises the Central Government on framing policies and measures for conservation of wildlife in the country and carries out impact assessment of projects in areas inhabited by Wildlife. It is an apex body to review all wildlife related matters and approve projects in and around National Parks & Wildlife Sanctuaries. The National Board for Wildlife is chaired by the prime minister.
- Constitution of the National Board for Wild Life.— The Central Government shall, within three months from the date of commencement of the Wild Life (Protection) Amendment Act, 2002 (16 of 2003), constitute the National Board for Wild Life consisting of the following members, namely:—
 - (a) the Prime Minister as Chairperson;
 - (b) the Minister in-charge of Forests and Wildlife as Vice-Chairperson;
 - (c) three members of Parliament of whom two shall be from the House of the People and one from the Council of States;
 - (d) Member, Planning Commission in-charge of Forests and Wild Life;
 - (e) five persons to represent non-governmental organizations to be nominated by the Central Government;
 - (f) ten persons to be nominated by the Central Government from amongst eminent conservationists, ecologists and environmentalists;
 - (g) the Secretary to the Government of India in-charge of the Ministry or Department of the Central Government dealing with Forests and Wild Life;
 - (h) the Chief of the Army Staff;

(i) the Secretary to the Government of India in-charge of the Ministry of Defence;

(j) the Secretary to the Government of India in-charge of the Ministry of Information and Broadcasting;

(k) the Secretary to the Government of India in-charge of the Department of Expenditure, Ministry of Finance;

(l) the Secretary to the Government of India, Ministry of Tribal Welfare;

(m) the Director-General of Forests in the Ministry or Department of the Central Government dealing with Forests and Wild Life;

(n) the Director-General of Tourism, Government of India;

(o) the Director-General, Indian Council for Forestry Research and Education, Dehradun;

(p) the Director, Wild Life Institute of India, Dehradun;

(q) the Director, Zoological Survey of India;

(r) the Director, Botanical Survey of India;

(s) the Director, Indian Veterinary Research Institute;

(t) the Member-Secretary, Central Zoo Authority;

(u) the Director, National Institute of Oceanography;

(v) one representative each from ten States and Union territories by rotation, to be nominated by the Central Government;

(w) the Director of Wild Life Preservation who shall be the Member-Secretary of the National Board.

a person or sample of bodily tissue. It helps in determining an individual's DNA characteristics.

Q.90) Ans: B

Exp.

- The report is jointly published by the World Wide Fund for Nature (WWF) and the United Nations Environment Programme (UNEP). It was released on July 8, 2021. The Title of the report A future for all – the need for human-wildlife coexistence. The report highlights that reducing human-wildlife conflict can benefit biodiversity, local communities, and society, contributing to sustainable development and production.

Source:

<https://www.unep.org/resources/report/future-all-need-human-wildlife-coexistence#:~:text=This%20joint%20report%20betw,een%20the,terrestrial%20and%20marine%20carnivor,e%20species>

Q.91) Ans: C

Exp.

- Pair 1 is incorrectly matched: Haliday Islands WLS is located in the state of West Bengal.
- Pair 2 is correctly matched: JorePokhri Salamander WLS is situated in the state of West Bengal.
- Pair 3 is incorrectly matched: Mahananda WLS is present in West Bengal.
- Pair 4 is incorrectly matched: Bori WLS is situated in Madhya Pradesh whereas Bor WLS is present in Maharashtra.
- Pair 5 is correctly matched: Gandhi Sagar WLS is situated in Madhya Pradesh.

Q.92) Ans: D

Exp.

- **Statement 1 is incorrect :** An initiative of the Khadi and Village Industries Commission (KVIC) under Project RE-HAB (Reducing Elephant-Human Attacks using Bees) intends to create "bee fences" to thwart elephant attacks in human habitations using honeybees.
- **Statement 2 is incorrect :** KVIC started implementing this programme by setting up 15 to 20 interspersed bee boxes at Nararhole National Park in Kodagu district of Karnataka. The bee boxes are set up in the passageways of

Q.89) Ans: B

Exp.

- Indian rhino could be the first wild animal species in India to have all its members DNA-sequenced. The Ministry of Environment Forest and Climate Change (MoEFCC) by 2021, all rhinos in the country. The project will include World Wide Fund for Nature-India (WWF-India) and the Centre-funded Wildlife Institute of India (WII). The exercise would be useful in curbing poaching and gathering evidence in wildlife crimes involving rhinos.
- DNA profiling is the process where a specific DNA pattern, called a profile, is obtained from



human-elephant conflict zones to block the entrance of elephants to human habitations.

- Bee boxes have been placed on the ground as well as hung from the trees. High resolution, night vision cameras have been installed at strategic points to record the impact of bees on elephants and their behaviour in these zones.
- The biggest advantage of Project RE-HAB is that it dissuades elephants without causing any harm to them. Besides being extremely cost-effective as compared to various other measures such as digging trenches or erecting fences.

Source:

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=1704902>

Q.93) Ans: B

Exp.

- The index evaluates states based on their attractiveness for rooftop development. The index has been designed by MNRE, Shakti Sustainable Energy Foundation, Associated Chambers of Commerce and Industry of India and Ernst & Young.

The Ministry of New and Renewable Energy (MNRE) has set a target of 175 GW of renewable energy capacity by 2022, of which 100 GW solar power is to be operational by March 2022. Under which, 40 GW is expected to come from grid connected solar rooftops.

- The index captures five key aspects namely
(a) Robustness of policy framework
(b) Implementation environment
(c) Investment climate
(d) Consumer experience and
(e) Business ecosystem
- The states which have scored A++ for the financial year 2018-19 are as follows:
 - Karnataka - SARAL score of 78.8
 - Telangana - SARAL score of 72.2
 - Gujarat - SARAL score of 67.9
 - Andhra Pradesh - SARAL score of 66.1whereas

- Manipur, Tripura, Meghalaya and Jammu & Kashmir(former) constituted the bottom 4 of the list.

Q.94) Ans: D

Exp.

- **Statement 1 is correct :** In a major push towards deep technology and driving the country to become a digitally transformed nation, Atal Innovation Mission (AIM), NITI Aayog today launched AIM-PRIME (Program for Researchers on Innovations, Market-Readiness & Entrepreneurship), an initiative to promote and support science-based deep-tech startups & ventures across India.
- **Statement 2 is correct :** AIM has joined hands with Bill & Melinda Gates Foundation (BMGF) to launch this nationwide program which will be implemented by Venture Center - a non-profit technology business incubator.
- **Statement 3 is correct :** The benefits of this program are aimed at addressing specific issues through training and guidance over a period of 12 months. Candidates selected for the program will get access to in-depth learning via a comprehensive lecture series, live team projects, exercises, and project-specific mentoring. The AIM-PRIME program is specifically tailored for the rapid scaling up of deep-tech science ventures in India, providing not just the necessary intellect and support but also the exposure they rightly deserve.

Source:

<https://pib.gov.in/PressReleaseFramePage.aspx?PRID=1708798>

Q.95) Ans: D

Exp.

- **Statement 1 is incorrect :** The rocket system has been developed jointly by Pune based Armament Research and Development Establishment (ARDE) and High Energy Materials Research Laboratory (HEMRL) with manufacturing support from M/s Economic Explosives Limited, Nagpur. Recently the Defence Research and Development Organisation (DRDO) successfully test fired an extended range

version of indigenously developed Pinaka rocket from a Multi-Barrel Rocket Launcher (MBRL) at Integrated Test Range (ITR), Chandipur off the coast of Odisha.

- **Statement 2 is incorrect** : Pinaka Rocket System is a multibarrel multiple rocket launcher of the Indian Army that can destroy targets at distances up to 45 kms.
- A ramjet, sometimes referred to as a flying stovepipe or an athodyd (aero thermodynamic duct), is a form of airbreathing jet engine that uses the engine's forward motion to compress incoming air without an axial compressor or a centrifugal compressor. Because ramjets cannot produce thrust at zero airspeed, they cannot move an aircraft from a standstill. A ramjet-powered vehicle, therefore, requires an assisted take-off like a rocket assist to accelerate it to a speed where it begins to produce thrust.
- Ramjets work most efficiently at **supersonic speeds around Mach 3** (three times the speed of sound) and can operate up to speeds of Mach 6. However, the ramjet efficiency starts to drop when the vehicle reaches hypersonic speeds.
- A **scramjet** (supersonic combustion **ramjet**) is a variant of a **ramjet** airbreathing jet engine in which combustion takes place in supersonic airflow. Scramjet powered vehicles are envisioned to operate at speeds up to **at least Mach 15**.

Q.96) Ans: C

Exp.

- A special event on 'Ambition to Impact: Opportunities for Global Collaboration in India's Clean Energy Economy' at the United Nations High-Level Dialogue on Energy (HLDE) 2021 was co-organised by the Ministry of New and Renewable Energy, Government of India; the Permanent Mission of India to the UN at New York, and the Council on Energy, Environment and Water (CEEW).
- It showcased India's efforts to achieve the 2030 Sustainable Development Agenda through a citizen-centric energy transition and focused on enhancing the speed of India's clean energy deployment through global collaboration. It discussed the multilateral

efforts required for this transition through conducive international policies, co-development of technologies, pooled finances for demonstrations, and investments to create markets and scale up deployment.

Q.97) Ans: D

Exp.

- Shri R.K. Singh, Minister of State (I/C) for Power and New & Renewable Energy launched "The India Story" booklet on Indian initiatives shaping energy transition.
- The India Story" booklet captures the essence of some of the flagship initiatives that have accelerated energy transition. These will continue to power our ambitious renewable energy programmes, with the end goal of ensuring access to affordable, reliable, sustainable and modern energy for all, while always keeping the citizen at the center of this transition

Q.98) Ans: C

Exp.

- **Statement 1 is correct** : One Nation One Standard mission focuses on setting global benchmarks with respect to quality standards of Indian goods. To attain the One Nation One Standard vision of the Government of India, BIS(Bureau of Indian Standards) launched a scheme which provides for Recognition of SDO. Through this scheme, BIS aims at aggregating and integrating the existing capabilities and dedicated domain specific expertise available with various organizations in the country engaged in standards development in their specific sectors, and enable convergence of all standard development activities in the country resulting in "One National Standard for One Subject" The recognition will be valid for 3 years and will require renewal after completion of the validity period.
- **Statement 2 is correct** : RDSO (Research Design & Standards Organization) of Indian Railways has become the FIRST Institution to be declared SDO under "One Nation One



Standard" mission on BIS (Bureau of Indian Standards) which is Institution under Department of Consumer Affairs.

Research Designs & Standards Organization (RDSO), Lucknow, which is the sole R&D Wing of the Ministry of Railways, is one of India's leading Standard formulating bodies undertaking standardization work for the railway sector.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1723360>

Q.99) Ans: B

Exp.

- The Union Cabinet approved a MoU between the Ministry of Mines, Government of India and the Secretariat of Mining Policy of the Ministry of Productive Development of the Argentine Republic.

The objectives of the MoU are to strengthen the activities involved like cooperation for encouraging minerals exploration and development,

- including extraction, mining and beneficiation of lithium;
- possibilities of forming joint venture in the field of base metals, critical and strategic minerals for mutual benefit;
- exchange of technical and scientific information and interchange of ideas and knowledge; training and capacity building; and
- promotion of investment and development in the area of mining activities, would serve the objective of innovation

Source: <https://pib.gov.in/PressReleasePage.aspx?PRID=1723641>

widely in cellular phones and notebook computers. Some of the applications of li-ion batteries are:

- Emergency Power Backup Or UPS
- Dependable Electric And Recreational Vehicle Power
- Solar Power Storage
- Mobile, Laptops, and other commonly used consumer electronic goods
- Surveillance Or Alarm Systems In Remote Locations
- Energy Storage Systems
- Smartwatches
- The various Advantages of Lithium-Ion Batteries over its peers are :
 - High Energy Density
 - Low Self Discharge
 - Low to Minimum Maintenance
 - Comes in all shapes and sizes

Extra edge by ONLYIAS:

- India's **first Lithium refinery will be set up in Gujarat.**
 - It will process Lithium ore to **produce battery-grade material.**
 - Lithium is a rare element not usually found in India, which would be **imported from Australia and processed in this refinery.**

Q.100) Ans: D

Exp:

All the Statements are correct:

- **Lithium-ion Batteries**
 - A lithium-ion battery or Li-ion battery is a type of rechargeable battery. Developed by Nobel Laureate Akira Yoshino, it is used